

E17304_265

The equation $-x^4 + x^2 + x + y^3 - y + 3 = 0$ has no integer solutions

Size $h = 35$, length $l = 12.58$

Theorem 1. *The equation*

$$-x^4 + x^2 + x + y^3 - y + 3 = 0 \tag{1}$$

has no integer solutions.

The proof uses the quartic Hilbert reciprocity law over the field $K = \mathbb{Q}(i)$. To a hypothetical integer solution we attach the values of 47 auxiliary polynomials $H_0, \dots, H_{46}$ with coefficients in $\mathbb{Z}[i]$, constants $c_0, \dots, c_{46} \in \mathbb{Z}[i]$, and a sum B of quartic Hilbert symbols of these values, recorded by a graph with 220 weighted edges. By the reciprocity law, the local contributions B_p of all primes p add up to 0 modulo 4. We show that $B_p = 0$ for $p \notin S$, compute B_p for the primes $p \in S$, and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the quartic Hilbert symbol at the places not above 2, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

1 Quartic Hilbert symbols over $\mathbb{Q}(i)$

The field. Let $K = \mathbb{Q}(i)$, where $i^2 = -1$. Its ring of integers is $\mathbb{Z}[i]$, and every element of K can be written uniquely as $A + Bi$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + Bi,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$. The units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[i]$. A prime $p \equiv 3 \pmod{4}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[i]/p\mathbb{Z}[i]$ with p^2 elements. If $p \equiv 1 \pmod{4}$, the polynomial $t^2 + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which i is the root of $t^2 + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and i reduces to r . The prime 2 is ramified: $\pi = 1 + i$ generates the unique prime ideal above 2, $\pi^2 = 2i$, and $\mathbb{Z}[i]/\pi\mathbb{Z}[i] = \mathbb{F}_2$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the quartic Hilbert symbol $(a, b)_v$ is a fourth root of unity, see [1, Chapter V, §3]. We write $(a, b)_v = i^{\beta_v(a, b)}$ with $\beta_v(a, b) \in \mathbb{Z}/4\mathbb{Z}$, and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$, $\beta_v(a, b) = -\beta_v(b, a)$, and $\beta_v(a, c^4) = 0$. In particular, $\beta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 4}$.

(H2) Let v be a finite place not above 2, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/4\mathbb{Z}$ by $\bar{u}^{(q-1)/4} = \bar{i}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\beta_v(a, b) = e\chi(w) - f\chi(u) + ef\chi(-1)$, and $\chi(-1) = (q-1)/2 \pmod{4}$. In particular, $\beta_v(a, b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\beta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \beta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\beta_v(a, b) = 0$.

The value of $\chi(-1)$ follows from $-1 = \bar{i}^2$ and $(-1)^{(q-1)/4} = \bar{i}^{(q-1)/2}$. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same.

Classes at the places not above 2. Let $p \neq 2$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $c_v(a) = (e \pmod{4}, \chi(u)) \in (\mathbb{Z}/4\mathbb{Z})^2$. By (H2) and (H1), c_v is additive, $c_v(a)$ determines the class of a modulo fourth powers, and $\beta_v(a, b) = e\chi(w) - f\chi(u) + \varepsilon_v ef$ for $c_v(a) = (e, \chi(u))$ and $c_v(b) = (f, \chi(w))$, where $\varepsilon_v = \chi(-1)$. We put $c_p(a) = c_v(a) \in (\mathbb{Z}/4\mathbb{Z})^2$ if $p \equiv 3 \pmod{4}$; then $q = p^2 \equiv 1 \pmod{8}$ and $\varepsilon_v = 0$. If $p \equiv 1 \pmod{4}$, we put $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in (\mathbb{Z}/4\mathbb{Z})^4$, where $r < r'$ are the roots of $t^2 + 1$ in $\{1, \dots, p-1\}$; here $q = p$, and $\varepsilon_v = 0$ if $p \equiv 1 \pmod{8}$ and $\varepsilon_v = 2$ if $p \equiv 5 \pmod{8}$. Accordingly, β_p is the sum of the forms β_v over the places v above p . For $p \equiv 3 \pmod{4}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/4} = (\bar{u}^{p-1})^{(p+1)/4} = 1$ and $\chi(u) = 0$.

The place above 2.

Lemma 2. *Let v be the place above 2. Every non-zero element of K_v is equal to $\pi^a i^b (1 + 2i)^{c5^d}$ times a fourth power, for unique $a, b, c, d \in \mathbb{Z}/4\mathbb{Z}$, and every unit congruent to 1 modulo π^7 is a fourth power. Put $c_2(A) = (a, b, c, d) \in (\mathbb{Z}/4\mathbb{Z})^4$. Then c_2 is additive, and $\beta_v(A, B) = c_2(A)^\top M_2 c_2(B)$, where*

$$M_2 = \begin{pmatrix} 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 3 & 2 & 0 & 0 \end{pmatrix} \pmod{4}.$$

Proof. For $\alpha \in \mathbb{Z}[i]$, the polynomial $g(s) = s - 3(1-i)s^2 - 8is^3 + 4(1+i)s^4 + \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[i]$. Since $(1 + \pi^3 s)^4 = 1 - \pi^7(s - 3(1-i)s^2 - 8is^3 + 4(1+i)s^4)$, as one checks using $\pi^5 = -4 - 4i$, $\pi^6 = -8i$ and $\pi^7 = 8 - 8i$, we obtain $(1 + \pi^3 s)^4 = 1 + \pi^7 \alpha$. Hence every unit congruent to 1 modulo π^7 is a fourth power in K_v , and the class of a unit modulo fourth powers depends only on its residue modulo π^7 . The ring $\mathbb{Z}[i]/\pi^7 \mathbb{Z}[i]$ has 2^7 elements, 64 of which are units, and a direct enumeration shows that the 64 products $i^b (1 + 2i)^{c5^d}$ with $b, c, d \in \{0, 1, 2, 3\}$ are pairwise distinct modulo π^7 . Hence they represent all units modulo π^7 , the group of these units has exponent 4, and the fourth power of every unit is congruent to 1 modulo π^7 . Therefore two units are in the same class modulo fourth powers if and only if they are congruent modulo π^7 , and the 64 products represent the 64 classes of units. Since π^4 is a fourth power, the exponent of π is the valuation modulo 4, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements $\pi, i, 1 + 2i$ and 5. Their norms are 2, 1, 5 and 25, hence, for each such pair, all the symbols at the places not above 2 and 5 vanish by (H2), and, by (H3), β_v at the place above 2 is minus the sum of the values at the two places above 5. These places correspond to $\bar{i} = 2$ and $\bar{i} = 3$ in $\mathbb{F}_5$; there 5 is a uniformizer, $q = 5$, $\chi(u)$ is defined by $\bar{u} = \bar{i}^{\chi(u)}$, and $\chi(-1) = 2$. The classes (e, χ) of the four

elements at these places are

	$\bar{i} = 2$	$\bar{i} = 3$
π	$(0, 3)$	$(0, 2)$
i	$(0, 1)$	$(0, 1)$
$1 + 2i$	$(1, 3)$	$(0, 3)$
5	$(1, 0)$	$(1, 0)$

where, for $1 + 2i$ at the place with $\bar{i} = 2$, the unit part is $(1 + 2i)/5 \equiv (1 + 2 \cdot 7)/5 = 3 \pmod{5}$, using $i \equiv 7 \pmod{25}$. Formula (H2) and the sum over these two places give M_2 . $\square$

We write β_2 for the form of Lemma 2. Since $2 = i^3\pi^2$, we have $c_2(2) = (2, 3, 0, 0)$; also $c_2(-1) = (0, 2, 0, 0)$ and $c_2(5) = (0, 0, 0, 1)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 3. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[i]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $c_p(z) \in b + D$ for the vector b and the subgroup D given as follows.*

- (i) $p \equiv 3 \pmod{4}$. If $\zeta \neq 0$, then $b = c_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $(\mathbb{Z}/4\mathbb{Z})^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{4}$. For each root r , let ρ be the root of $t^2 + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $c_{v_r}(z)$ are those of $c_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 2$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = c_2(\zeta)$, and $D = 0$ if $m \geq 7$, while $D = D_m$ if $m \leq 6$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, and $(0, 0, 0, 1)$; D_2 is spanned by $(0, 2, 0, 0)$, $(0, 0, 1, 0)$, and $(0, 0, 0, 1)$; D_3 is spanned by $(0, 2, 1, 0)$, $(0, 0, 2, 0)$, and $(0, 0, 0, 1)$; D_4 is spanned by $(0, 0, 2, 0)$ and $(0, 0, 0, 1)$; D_5 is spanned by $(0, 0, 2, 0)$ and $(0, 0, 0, 2)$; D_6 is spanned by $(0, 0, 0, 2)$. If $\zeta = 0$, then $b = 0$ and $D = (\mathbb{Z}/4\mathbb{Z})^4$.
- (iv) $p = 2$, $z \in \mathbb{Z}$, so that $B = 0$. If $2^k \nmid A$, let $j = v_2(A)$ and $u = A/2^j$. Then $b = c_2(A)$ and $D = 0$ if $k - j \geq 4$, while $b = j c_2(2) + c_2(u)$ and $D = R_{k-j}$ if $k - j \leq 3$, where R_1 is spanned by $(0, 2, 0, 0)$ and $(0, 0, 0, 1)$; R_2 is spanned by $(0, 0, 0, 1)$; R_3 is spanned by $(0, 0, 0, 2)$. If $2^k \mid A$, then $b = 0$ and D is spanned by $c_2(2)$, $c_2(-1)$ and $c_2(5)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'i$ and $A + Bi$ with $i \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 2, the class of a unit depends only on its residue modulo π^7 ; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo π^7 shows that their classes form D_m . (iv) If $2^k \nmid A$, then $z = 2^j u'$ with a rational unit $u' \equiv u \pmod{2^{k-j}}$, and u'/u is a rational unit congruent to 1 modulo 2^{k-j} . A rational unit congruent to 1 modulo 16 is congruent to 1 modulo π^8 , hence a fourth power, and the odd residues modulo 16 are $\pm 5^s$; hence the classes of the rational units congruent to 1 modulo 2^{k-j} form R_{k-j} , and this group is 0 if $k - j \geq 4$. If $2^k \mid A$, then also the valuation j is unknown, and every non-zero rational number is $2^j(\pm 5^s)$ times a fourth power. $\square$

Lemma 4. *Let p be a prime, and consider vertices i with vectors b_i , subgroups D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2, 3\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$*

for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then

$$\sum_{(i,j)} w_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in generating sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and $\beta_p(u, v) = -\beta_p(v, u)$, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 5. Let p be an odd prime and v a place above p , and let τ denote complex conjugation. Let $\tau(v)$ be the place of the conjugate prime ideal, so that $\tau(v) = v$ if $p \equiv 3 \pmod{4}$ and $\{v, \tau(v)\} = \{v_r, v_{r'}\}$ if $p \equiv 1 \pmod{4}$. Then $v(\tau(a)) = \tau(v)(a)$ and $c_v(\tau(a)) = \sigma(c_{\tau(v)}(a))$ for $a \in K^\times$, where $\sigma(e, \chi) = (e, -\chi)$. In particular, $c_{\tau(v)}(a) = \sigma(c_v(a))$ for $a \in \mathbb{Q}^\times$.

Proof. The automorphism τ maps the prime ideal of $\tau(v)$ onto that of v , so $v(\tau(a)) = \tau(v)(a)$, and it induces an isomorphism from the residue field of $\tau(v)$ onto that of v , which maps $\bar{i}$ to $-\bar{i} = \bar{i}^3$; both residue fields have q elements. Write $a = p^e u$ with a unit u at $\tau(v)$, so that $\tau(a) = p^e \tau(u)$. Applying τ to $\bar{u}^{(q-1)/4} = \bar{i}^{\chi(u)}$ gives $\overline{\tau(u)}^{(q-1)/4} = \bar{i}^{-\chi(u)}$ in the residue field of v . Hence $c_v(\tau(a)) = (e, -\chi(u)) = \sigma(c_{\tau(v)}(a))$. For $a \in \mathbb{Q}^\times$, apply this to $a = \tau(a)$ and the place $\tau(v)$, and use $\tau(\tau(v)) = v$ and $\sigma^2 = 1$. $\square$

2 The graph

Put $F(x, y) = -x^4 + x^2 + x + y^3 - y + 3$, so that (1) is the equation $F(x, y) = 0$. The certificate consists of 47 polynomials H_i with coefficients in $\mathbb{Z}[i]$ and constants $c_i \in \mathbb{Z}[i]$, listed in Table 1, and of 220 edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2, 3\}$, listed below. For an edge (i, j) we put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij} \pmod{4}$, and $\rho_{ij} = 0$ if i and j are not joined by an edge. The edges $(i, j; w_{ij})$ are (0, 8; 1), (0, 9; 3), (0, 13; 3), (0, 21; 1), (0, 33; 2), (0, 37; 3), (0, 39; 2), (1, 8; 2), (1, 9; 3), (1, 14; 1), (1, 21; 3), (1, 31; 3), (1, 34; 2), (2, 9; 2), (2, 20; 3), (2, 36; 3), (2, 40; 3), (3, 8; 1), (3, 19; 2), (3, 21; 3), (3, 30; 2), (3, 37; 2), (3, 39; 2), (4, 12; 1), (4, 14; 1), (4, 16; 2), (4, 21; 1), (4, 34; 1), (4, 36; 1), (4, 37; 1), (4, 40; 1), (5, 13; 2), (5, 14; 2), (5, 16; 1), (5, 17; 3), (5, 26; 3), (5, 28; 2), (5, 29; 2), (5, 31; 3), (5, 32; 1), (5, 34; 2), (5, 43; 1), (6, 7; 3), (6, 16; 3), (6, 34; 3), (6, 36; 1), (6, 40; 1), (7, 12; 1), (7, 17; 2), (7, 20; 2), (7, 21; 2), (7, 37; 3), (7, 38; 2), (8, 9; 1), (8, 16; 2), (8, 25; 3), (8, 26; 2), (8, 30; 3), (8, 32; 1), (8, 33; 2), (8, 37; 2), (8, 42; 3), (8, 44; 3), (8, 46; 2), (9, 11; 3), (9, 12; 2), (9, 19; 3), (9, 21; 2), (9, 23; 3), (9, 25; 1), (9, 28; 2), (9, 30; 1), (9, 33; 2), (9, 34; 1), (9, 39; 2), (9, 40; 2), (9, 44; 2), (9, 45; 2), (9, 46; 1), (10, 11; 3), (10, 12; 1), (10, 13; 3), (10, 19; 3), (10, 26; 1), (10, 37; 1), (10, 39; 1), (11, 26; 1), (11, 28; 2), (11, 29; 1), (11, 35; 3), (12, 14; 2), (12, 16; 1), (12, 21; 1), (12, 25; 1), (12, 28; 2), (12, 31; 1), (12, 39; 1), (12, 40; 3), (12, 43; 3), (12, 46; 1), (13, 16; 2), (13, 24; 2), (13, 31; 3), (13, 38; 3), (13, 40; 3), (13, 44; 3), (14, 22; 1), (14, 23; 1), (14, 33; 1), (14, 34; 2), (14, 36; 3), (14, 44; 2), (15, 17; 3), (15, 20; 2), (15, 23; 1), (15, 25; 1), (15, 26; 1), (15, 29; 2), (15, 37; 2), (15, 43; 3), (16, 22; 3), (16, 24; 2), (16, 27; 1), (16, 38; 3), (16, 43; 3), (16, 44; 1), (16, 46; 2), (17, 23; 1), (17, 32; 3), (17, 40; 2), (17, 43; 3), (17, 46; 1), (18, 23; 3), (18, 25; 1), (18, 31; 3), (18, 38; 3), (18, 40; 1), (18, 45; 1), (19, 20; 1), (19, 25; 1), (19, 26; 3), (19, 27; 2), (19, 30; 1), (19, 31; 2), (19, 38; 3), (19, 39; 2), (19, 41; 1), (19, 45; 3), (20, 33; 1), (20, 38; 3), (20, 45; 3), (21, 25; 3), (21, 27; 1), (21, 32; 2), (21, 34; 3), (21, 38; 3), (21, 39; 1), (21, 43; 1), (21, 44; 2), (21, 45; 2), (22, 24; 2), (22, 25; 3), (22, 26; 1), (22, 39; 3), (23, 24; 2), (23, 31; 1), (23, 36; 3), (23, 44; 1), (23, 45; 3), (23, 46; 3), (24, 35; 2), (24, 39; 2), (25, 26; 1), (25, 32; 1), (25, 37; 3), (25, 38; 1), (25, 40; 1), (25, 41; 1), (25, 45; 1), (26, 31; 1), (26, 34; 1), (26, 37; 1), (26, 40; 2), (26, 41; 1), (26, 45; 3), (27, 28; 2), (27, 29; 2), (27, 30; 3), (27, 31; 1), (27, 38; 1), (27, 40; 3), (27, 46; 2), (28, 30; 2), (28, 33; 2), (29, 30; 2), (29, 33; 1),

(29, 34; 1), (29, 38; 3), (29, 39; 3), (29, 46; 3), (30, 44; 1), (31, 37; 3), (31, 45; 2), (31, 46; 1), (33, 34; 2), (33, 35; 3), (33, 36; 2), (33, 41; 2), (33, 44; 1), (34, 36; 3), (34, 37; 3), (34, 45; 1), (36, 37; 2), (36, 46; 2), (37, 40; 2), (37, 44; 2), (38, 40; 3), (39, 40; 1), (39, 41; 1), (41, 43; 1). Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $H_i = \varepsilon_i(\lambda_i \eta - s_i(\xi))$ for a unit ε_i of $\mathbb{Z}[i]$, an element $\lambda_i \in \mathbb{Z}[i]$ whose norm is a power of 2, and a polynomial s_i with coefficients in $\mathbb{Z}[i]$; the last column of Table 1 gives $\lambda_i \eta = s_i(\xi)$ for the linear vertices, where $\lambda_i = 1$ if no factor is shown. For a linear vertex i , we put $f_i(t) = \lambda_i^d F$ and $g_{ij}(t) = \lambda_i^{d_j} H_j$ evaluated at $\xi = t$, $\eta = s_i(t)/\lambda_i$, where d and d_j are the degrees of F and H_j in η ; these are polynomials in t with coefficients in $\mathbb{Z}[i]$. Put $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$, where $\rho_{ij} \in \{0, 1, 2, 3\}$. We write τ for complex conjugation, and for a polynomial P with coefficients in $\mathbb{Z}[i]$ we write $\tau(P)$ for the polynomial obtained by applying τ to every coefficient. Every vertex i listed in the first column of Table 2 has a *factor* φ_i , given in the table, of one of the following three kinds: $\varphi_i = 1$; or φ_i is a polynomial with integer coefficients, and then H_i also has integer coefficients; or $\varphi_i = \tau(\Pi_{i^*})$ for a vertex i^* with $H_{i^*} = \tau(H_i)$. The vertices i^* of the last kind are not listed in the first column, and every vertex is either listed there or equal to i^* for exactly one listed vertex i . For every linear vertex i in Table 2, the table gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[i]$ such that

$$D_i^4 \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - \varphi_i(t) w_i^4 = f_i Q_i \quad (2)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[i]$, where $\varphi_i(t)$ is φ_i evaluated at $\xi = t$, $\eta = r_i(t)$; the table also gives an integer $e_i \geq 0$ with $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod{4}$, and $e_i = 0$ if $\lambda_i = 1$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For every other vertex i in this table, Table 2 gives an integer $D_i \geq 1$ and polynomials W_i , A_i and B_i in x and y with coefficients in $\mathbb{Z}[i]$ such that

$$D_i^4 c_i \prod_j H_j^{\rho_{ij}} - \varphi_i W_i^4 = A_i F + B_i H_i. \quad (3)$$

For the edges (i, j) listed in Table 3, the table gives a linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $i^2 = -1$.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) , which lie in $\mathbb{Z}[i]$.

Step 1: the values are non-zero. For $i \in \{1, 7, 9, 10, 11, 14, 15, 16, 18, 21, 22, 24, 26, 31, 32, 33, 34, 36, 37, 39, 40, 41, 42, 44\}$, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{8, 12, 13, 17, 19, 23, 25, 27, 28, 29, 30, 35, 38, 43, 45, 46\}$, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{0, 20\}$, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{2, 3, 5, 6\}$, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. For $i \in \{4\}$, there is no residue pair (a, b) modulo 13 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{13}$, as one checks for the 13^2 residue pairs; hence $H_i \neq 0$. Also, all c_i are non-zero.

Step 2: reciprocity. For every place v of K , put

$$B_v = \sum_{(i,j)} w_{ij} \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/4\mathbb{Z}, \quad (4)$$

the first sum over the edges, and, for a rational prime p , put $B_p = \sum_{v|p} B_v$. Applying (H3) to every pair (H_i, H_j) with an edge, multiplied by w_{ij} , and to every pair (H_i, c_i) , and adding, we find that $B_v = 0$ for all but finitely many v and $\sum_p B_p = 0$, as the complex place contributes 0. By (H1) and the definition of c_p , $B_p = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$, where β_p is the form of Lemma 2 if $p = 2$.

Step 3: the primes outside S . Let $S = \{2, 3, 5, 7, 13, 17, 29, 31, 37, 41, 61, 109\}$. It contains 2 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides H_i and H_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $H_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$ and λ_k is a unit at v , we have $\eta \equiv s_k(t)/\lambda_k$ modulo v , hence $f_k(t) \equiv \lambda_k^d F(x, y) = 0$ and $g_{kl}(t) \equiv \lambda_k^{d_l} H_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = A f_k + C g_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[i]$, so v divides R_{ij} , a contradiction. By (H2), a term of (4) in which all arguments are units vanishes. Now let v divide H_i . Collecting the terms of (4) which contain H_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\beta_v(H_i, \Pi_i)$, and Π_i is a unit at v ; by (H2), $\beta_v(H_i, \Pi_i) = T(i, v)$, where $T(i, v) = v(H_i) \chi_v(\Pi_i)$. Hence B_p is the sum of $T(i, v)$ over the pairs (i, v) with $v | p$ and $v | H_i$. For a place v above p , let $\tau(v)$ be the place of the conjugate prime ideal; by Lemma 5, $v(\tau(a)) = \tau(v)(a)$ for $a \in \mathbb{Z}[i]$, and $\chi_v(\tau(a)) = -\chi_{\tau(v)}(a)$ if a is a unit at $\tau(v)$. Now let i be listed in Table 2, and let $v | H_i$. We claim that $\chi_v(\Pi_i) = \chi_v(\varphi_i)$. If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv \lambda_i^{d_j} H_j$, $\varphi_i(t) \equiv \varphi_i$ and $f_i(t) \equiv 0$ modulo v as above, and (2) gives $D_i^4 \lambda_i^{E_i} \Pi_i \equiv \varphi_i w_i(t)^4$ modulo v , where $E_i = e_i + \sum_j \rho_{ij} d_j$ is divisible by 4, and λ_i is a unit at v . Otherwise, (3) gives $D_i^4 \Pi_i \equiv \varphi_i W_i(x, y)^4$ modulo v . As $D_i^4 \lambda_i^{E_i} \Pi_i$ is a unit at v , so are φ_i and the fourth power, and the claim follows, as $\lambda_i^{E_i}$ is a fourth power. If $\varphi_i = 1$, then $T(i, v) = 0$. If φ_i has integer coefficients, then $H_i \in \mathbb{Z}$ and $\varphi_i \in \mathbb{Z}$, so $\tau(v) | H_i$ and $\tau(v)(H_i) = v(H_i)$; if $\tau(v) = v$, then $\chi_v(\varphi_i) = 0$ by Section 1, so $T(i, v) = 0$, and otherwise $T(i, v) + T(i, \tau(v)) = v(H_i)(\chi_v(\varphi_i) + \chi_{\tau(v)}(\varphi_i)) = 0$, as $\chi_v(\varphi_i) = \chi_v(\tau(\varphi_i)) = -\chi_{\tau(v)}(\varphi_i)$. If $\varphi_i = \tau(\Pi_{i^*})$, then $H_{i^*} = \tau(H_i)$, so $\tau(v) | H_{i^*}$ and $\tau(v)(H_{i^*}) = v(H_i)$, and $T(i, v) + T(i^*, \tau(v)) = v(H_i)(\chi_v(\tau(\Pi_{i^*})) + \chi_{\tau(v)}(\Pi_{i^*})) = 0$. This groups the pairs (i, v) into the singletons $\{(i, v)\}$ and the sets $\{(i, v), (i, \tau(v))\}$ and $\{(i, v), (i^*, \tau(v))\}$, and every pair (j, v) with $v | p$ and $v | H_j$ lies in exactly one of these groups. The terms of each group add up to 0; hence $B_p = 0$.

Step 4: the primes in S . Let $p \in S$. Here 3 is inert, the places above 5 are v_2 and v_3 , with $\varepsilon_v = 2$, 7 is inert, the places above 13 are v_5 and v_8 , with $\varepsilon_v = 2$, the places above 17 are v_4 and v_{13} , with $\varepsilon_v = 0$, the places above 29 are v_{12} and v_{17} , with $\varepsilon_v = 2$, 31 is inert, the places above 37 are v_6 and v_{31} , with $\varepsilon_v = 2$, the places above 41 are v_9 and v_{32} , with $\varepsilon_v = 0$, the places above 61 are v_{11} and v_{50} , with $\varepsilon_v = 2$, and the places above 109 are v_{33} and v_{76} , with $\varepsilon_v = 2$. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 3, applied with ζ the residue of $H_i(a, b)$, gives a vector b_i and a subgroup D_i with $c_p(H_i) \in b_i + D_i$; when H_i has rational coefficients, $H_i \in \mathbb{Z}$, and we use this in (i) and (iv). With $\gamma_i = c_p(c_i)$, if the hypotheses of Lemma 4 hold, then B_p is determined by a , b and k . If these hypotheses do not hold and p is odd, we use Lemma 5: $c_{\tau(v)}(H_i) = \sigma(c_v(H_i))$ for every vertex i with $H_i \in \mathbb{Z}$, and $c_{\tau(v)}(H_{i^*}) = \sigma(c_v(H_i))$ for the pairs (i, i^*) of Section 2, for each place v above p . Let U be the set of vectors $(c_i)_i$ with $c_i \in b_i + D_i$ that satisfy these relations; it contains the vector of the classes $c_p(H_i)$. If B_p , computed from $(c_i)_i$ in place of the classes, takes the same value for all elements of U , then it is determined by a , b and k . Starting from the solutions of (1) modulo p , we refine

a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses or the condition on U hold, or until no solution remains. Table 4 lists the resulting terminal classes and the value of B_p in each; every integer solution lies in one of them. For $p = 109$, every residue class modulo p that contains a solution of (1) modulo p is already terminal, with $B_{109} = 0$; these classes are not listed in Table 4. The result is $B_2 \in \{0, 2, 3\}$, $B_3 = 0$, $B_5 = 0$, $B_7 = 0$, $B_{13} = 3$, $B_{17} = 0$, $B_{29} = 0$, $B_{31} = 0$, $B_{37} = 0$, $B_{41} = 0$, $B_{61} = 0$, $B_{109} = 0$.

Step 5: contradiction. By Steps 2 and 3, $\sum_{p \in S} B_p = 0$. But by Step 4, $\sum_{p \in S} B_p \in \{1, 2, 3\}$ in $\mathbb{Z}/4\mathbb{Z}$. This contradiction proves the theorem. $\square$

References

- [1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$. Here $\langle A, B \rangle = A + Bi$.

i	H_i	c_i	chart
0	$\langle -x + y - 1, 0 \rangle$	$\langle 1, 1 \rangle$	$y = \langle x + 1, 0 \rangle$
1	$\langle -x - 2, 0 \rangle$	$\langle 3, 5 \rangle$	$x = \langle -2, 0 \rangle$
2	$\langle -x - y + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, 0 \rangle$
3	$\langle -x + y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, 0 \rangle$
4	$\langle -x^2 + y - 1, 0 \rangle$	$\langle 0, 2 \rangle$	$y = \langle x^2 + 1, 0 \rangle$
5	$\langle x^2 - x + y, 0 \rangle$	$\langle 0, 2 \rangle$	$y = \langle -x^2 + x, 0 \rangle$
6	$\langle -x^2 + y, 0 \rangle$	$\langle -2, 2 \rangle$	$y = \langle x^2, 0 \rangle$
7	$\langle -x - y - 1, -1 \rangle$	$\langle 0, 1 \rangle$	$y = \langle -x - 1, -1 \rangle$
8	$\langle -2x + y - 1, -y - 1 \rangle$	$\langle -1, -1 \rangle$	$\langle 1, 1 \rangle y = \langle -1, 2x + 1 \rangle$
9	$\langle 2y - 1, -1 \rangle$	$\langle 135, 27 \rangle$	—
10	$\langle 2y - 1, 1 \rangle$	$\langle 1, 0 \rangle$	$2y = \langle 1, -1 \rangle$
11	$\langle -2x - 1, 1 \rangle$	$\langle 2, 0 \rangle$	$2x = \langle -1, 1 \rangle$
12	$\langle -x - y - 1, y \rangle$	$\langle 1, -1 \rangle$	$x = \langle -y - 1, y \rangle$
13	$\langle -x - 1, -y \rangle$	$\langle -1, 0 \rangle$	$y = \langle 0, x + 1 \rangle$
14	$\langle -x - y, -1 \rangle$	$\langle 0, 2 \rangle$	$y = \langle -x, -1 \rangle$
15	$\langle -x - y, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, 1 \rangle$
16	$\langle y, 1 \rangle$	$\langle -2, -1 \rangle$	$y = \langle 0, -1 \rangle$
17	$\langle -x + y, y - 1 \rangle$	$\langle -2, 2 \rangle$	$x = \langle y, y - 1 \rangle$
18	$\langle -x - y + 1, -1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, -1 \rangle$
19	$\langle -x + 1, y \rangle$	$\langle 1, -1 \rangle$	$y = \langle 0, -x + 1 \rangle$
20	$\langle -2x - y, y \rangle$	$\langle 0, 1 \rangle$	$\langle 1, 1 \rangle y = \langle 0, -2x \rangle$
21	$\langle -x, -y \rangle$	$\langle 1, -1 \rangle$	$y = \langle 0, x \rangle$
22	$\langle -x, y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x \rangle$
23	$\langle x^2 + x + y - 1, x + 1 \rangle$	$\langle -6, -8 \rangle$	$y = \langle -x^2 - x + 1, -x - 1 \rangle$
24	$\langle x^2 - x + y - 1, 1 \rangle$	$\langle 3, 4 \rangle$	$y = \langle -x^2 + x + 1, -1 \rangle$
25	$\langle x^2 + x + y, x + 1 \rangle$	$\langle 2, -2 \rangle$	$y = \langle -x^2 - x, -x - 1 \rangle$
26	$\langle x^2 + y, -x \rangle$	$\langle 2, 2 \rangle$	$y = \langle -x^2, x \rangle$
27	$\langle x + y - 1, x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, -x + 1 \rangle$
28	$\langle -x + y - 1, x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, -x + 1 \rangle$
29	$\langle y - 1, -2 \rangle$	$\langle 5, -1 \rangle$	$y = \langle 1, 2 \rangle$
30	$\langle -x - 1, -2 \rangle$	$\langle 27, -27 \rangle$	$x = \langle -1, -2 \rangle$
31	$\langle y + 1, -1 \rangle$	$\langle -243, 351 \rangle$	$y = \langle -1, 1 \rangle$
32	$\langle -2x + 1, -1 \rangle$	$\langle 6, 6 \rangle$	—
33	$\langle -x + 2, -1 \rangle$	$\langle 90, 54 \rangle$	$x = \langle 2, -1 \rangle$
34	$\langle -x + y + 1, -x^2 + x + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x - 1, x^2 - x - 1 \rangle$
35	$\langle x + y - 1, -x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, x - 1 \rangle$
36	$\langle y - 1, 1 \rangle$	$\langle 0, 1 \rangle$	$y = \langle 1, -1 \rangle$
37	$\langle -x - 1, 1 \rangle$	$\langle 18, 18 \rangle$	$x = \langle -1, 1 \rangle$
38	$\langle -x - 1, 2 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 2 \rangle$
39	$\langle -x, 1 \rangle$	$\langle 1, -1 \rangle$	$x = \langle 0, 1 \rangle$
40	$\langle -x, y - 1 \rangle$	$\langle -1, 1 \rangle$	$y = \langle 1, -x \rangle$
41	$\langle y + 1, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -1, -1 \rangle$
42	$\langle -2x + 1, 1 \rangle$	$\langle 1, 0 \rangle$	$2x = \langle 1, 1 \rangle$
43	$\langle -x + 1, 2 \rangle$	$\langle -2, 2 \rangle$	$x = \langle 1, 2 \rangle$
44	$\langle -x + y + 1, x^2 - x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x - 1, -x^2 + x + 1 \rangle$
45	$\langle -x - 1, 0 \rangle$	$\langle 0, 2 \rangle$	$x = \langle -1, 0 \rangle$
46	$\langle -x + 1, 0 \rangle$	$\langle 1, 1 \rangle$	$x = \langle 1, 0 \rangle$

Table 2: The data of the identities (2) and (3); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

$i = 0$	$D_i = 4$	φ_i below	ρ_{ij} : 8:1, 9:3, 13:3, 21:1, 33:2, 37:3, 39:2
φ_i		$-1336964032x^3 - 2510743488x^2 - 1877923264x - 1393665728$	
w_i		$\langle t^3 - 6t - 5, t^3 - 2t^2 - 4t + 5 \rangle$	

$i = 1$, $D_i = 407924916100$, φ_i below, ρ_{ij} : 8:2, 9:3, 14:1, 21:3, 31:3, 34:2
 φ_i $-26026968050153634315572224040000y^2 - 61716278011893890984961952980000y$
 $-121878576960961841061043469805000$
 w_i $\langle 67048t^2 + 162996t - 295936,$
 $-12716t^2 - 42772t + 378012 \rangle$

$i = 2$, $D_i = 2306217365240464$, φ_i below, ρ_{ij} : 9:2, 20:3, 36:3, 40:3
 φ_i $127507583431138720963959406339085568x^3 - 220324718764932870855212085757105664x^2$
 $+97424365569694349113155209516424448x - 138683363472929648869101714297446400$
 w_i $\langle -54487739t^3 + 18470420t^2 + 325079392t - 320461787,$
 $87734495t^3 + 140375192t^2 - 343549812t - 137604629 \rangle$

$i = 3$, $D_i = 239907958416$, φ_i below, ρ_{ij} : 8:1, 19:2, 21:3, 30:2, 37:2, 39:2
 φ_i $59724557779259756144325312x^3 + 56445367786652219601171456x^2 + 47846926134316315237960512x$
 $+93625653927744545002365312$
 w_i $\langle 96040t^3 + 249704t^2 - 633864t - 518616,$
 $268912t^3 - 345744t^2 - 403368t + 1094856 \rangle$

$i = 4$, $D_i = 36$, φ_i below, ρ_{ij} : 12:1, 14:1, 16:2, 21:1, 34:1, 36:1, 37:1, 40:1
 φ_i $-116640x^5 + 72576x^4 - 49248x^3 + 44064x^2 - 176904x + 83592$
 w_i $\langle 2t^5 - 6t^4 + 6t^3 - 10t^2 + 8t,$
 $-2t^5 - 2t^4 - 2t^3 - 10t^2 - 12 \rangle$

$i = 5$, $D_i = 128200622026202487612610953602500$, φ_i below, ρ_{ij} : 13:2, 14:2, 16:1, 17:3, 26:3, 28:2, 29:2, 31:3, 32:1, 34:2, 43:1
 φ_i $-1644030434761472272761372692875682115583879033889271640387555906328125000x^5$
 $+4591030104014586492466875377756885841861124648461906113661828489409375000x^4$
 $-1020758144781472714226095368379600666068812372135561846804080689865625000x^3$
 $-1693797514169277658715971470841169344642254802524519438404728628915625000x^2$
 $-769286679138995451119995242067503878407269028511692510014288771393750000x$
 $-3212140914516429795139681346593756462337503675836085745822408057568750000$
 w_i $\langle -2659121486045807t^5 + 8507083748386486t^4 - 9285120429451574t^3 - 8279653248337311t^2$
 $+11006861102918019t + 4086936284560807,$
 $-2613676180744501t^5 + 11104809198096148t^4 - 16836302567586532t^3 + 11802439508650577t^2$
 $+2205892274288617t - 1463754936729049 \rangle$

$i = 6$, $D_i = 15962134429696$, φ_i below, ρ_{ij} : 7:3, 16:3, 34:3, 36:1, 40:1
 φ_i $-1430838594542899303423148032x^5 + 1777317748543520811211816960x^4$
 $+92695053956354481112219648x^3 - 1457876344586979732480851968x^2$
 $+1123328023716838752535969792x - 2001577782890986187025022976$
 w_i $\langle -3687936t^5 - 5839232t^4 - 5224576t^3 - 4609920t^2 + 3687936t - 1843968,$
 $2458624t^5 + 5224576t^4 - 5839232t^3 - 8912512t^2 - 10449152t - 6761216 \rangle$

$i = 7$, $D_i = 1$, $\varphi_i = 1$, ρ_{ij} : 6:1, 12:1, 17:2, 20:2, 21:2, 37:3, 38:2
 w_i $\langle t^2 + t,$
 $3t^2 + 5t + 6 \rangle$

$i = 8$, $D_i = 8$, $\varphi_i = 1$, $e_i = 3$, ρ_{ij} : 0:3, 1:2, 3:3, 9:1, 16:2, 25:3, 26:2, 30:3, 32:1, 33:2, 37:2, 42:3, 44:3, 46:2
 w_i $\langle -2157184t^3 - 1721792t^2 + 2079616t + 1622208,$
 $-116864t^3 - 3186880t^2 - 2906368t + 1421376 \rangle$

$i = 9$, $D_i = 296$, $\varphi_i = \tau(\Pi_{10})$, ρ_{ij} : 0:1, 1:1, 2:2, 8:3, 11:3, 12:2, 19:3, 21:2, 23:3, 25:1, 28:2, 30:1, 33:2, 34:1, 39:2, 40:2, 44:2, 45:2, 46:1
 W $\langle -2172x^3 - 1560x^2 - 1962x - 6435,$
 $2508x^3 + 8400x^2 + 6432x + 5013 \rangle$

A $\langle 132651016519680x^{39} + 358157744603136x^{38} - 6526430012768256x^{37} - 30065352894185472x^{36}$
 $+27188483973414912x^{35} + 352801130742300672x^{34} + 387898931575701504x^{33}$
 $-1281325083625930752x^{32} - 3169138076399517696x^{31} + 330612553755611136x^{30}$
 $+8137302503940587520x^{29} + 7307357117647491072x^{28} - 5973080028166250496x^{27}$
 $-13966198597853641728x^{26} - 5698108772358672384x^{25} + 5454675399362254848x^{24}$
 $+8564048544788093952x^{23} + 6644192666980499712x^{22} - 641667345060925440x^{21}$
 $-8684797437068412672x^{20} - 2760096766965508224x^{19} + 9493769374330271616x^{18}$
 $+4478601572484170688x^{17} - 7043728300034370624x^{16} + 6439575435829272576x^{15}$
 $+41676153412278939552x^{14} + 74564719191296111232x^{13} + 95470202791094774112x^{12}$
 $+79291280671207793136x^{11} - 16321597453313532576x^{10} - 163486475351895539736x^9$
 $-245607743334010339800x^8 - 191077858985165518824x^7 - 65269375298533644084x^6$
 $+20946570641061555744x^5 + 35689657719961894932x^4 + 18497854116633082014x^3$
 $+4860470912897333034x^2 + 488998452352960908x - 29652317238729924,$
 $26530203303936x^{39} + 1313245063544832x^{38} + 4868292306272256x^{37} - 19360415861047296x^{36}$
 $-120810255157592064x^{35} - 42969809594990592x^{34} + 778183923311050752x^{33}$
 $+1334429430883024896x^{32} - 1401259013005639680x^{31} - 5759192907328278528x^{30}$
 $-2648941518902833152x^{29} + 8640434959032508416x^{28} + 11920800290982137856x^{27}$
 $-449074212302361600x^{26} - 11680334310861904896x^{25} - 9341932241540662272x^{24}$
 $-1824111228350260224x^{23} + 4507250086216855296x^{22} + 8113510032455985408x^{21}$
 $+1100041910202465792x^{20} - 10106845599760071552x^{19} - 5689381741462609920x^{18}$
 $+5169709556898244416x^{17} - 4938225006725076672x^{16} - 26383512934652383296x^{15}$
 $-26632123550790072480x^{14} + 610633846979223840x^{13} + 52888687041046806720x^{12}$
 $+135780155601549438096x^{11} + 208441339335247123440x^{10} + 183158840805837289992x^9$

$$\begin{aligned}
& +43662830571878150376x^8 - 101743013533443686208x^7 - 144259484662119451500x^6 \\
& - 93806231994466263900x^5 - 31326749835951665088x^4 - 1940627493070914654x^3 \\
& + 2673506728548778842x^2 + 1036703036341610244x + 118521276494569668) \\
B \quad & (-92855711563776x^{42} + 530604066078720x^{41}y - 3429028777033728x^{41} \\
& - 278567134691328x^{40}y^2 - 232139278909440x^{40}y - 4735641289752576x^{40} \\
& - 82906885324800x^{39}y^3 + 7740186813923328x^{39}y^2 - 49281510774767616x^{39}y \\
& + 91328566736093184x^{39} + 925240840224768x^{38}y^4 - 10119614422745088x^{38}y^3 \\
& + 26450612694024192x^{38}y^2 - 79079074429353984x^{38}y + 241147111999979520x^{38} \\
& - 1406100775108608x^{37}y^5 + 9678549792817152x^{37}y^4 + 17495010941239296x^{37}y^3 \\
& - 276039303757774848x^{37}y^2 + 799903869128441856x^{37}y - 659954144959193088x^{37} \\
& + 769375895814144x^{36}y^6 + 4624546063417344x^{36}y^5 - 83124101364350976x^{36}y^4 \\
& + 308325732109811712x^{36}y^3 - 749013135709519872x^{36}y^2 + 1906630368535756800x^{36}y \\
& - 2746358145495539712x^{36} + 497441311948800x^{35}y^7 - 20240886983196672x^{35}y^6 \\
& + 147630632560164864x^{35}y^5 - 334005310770315264x^{35}y^4 - 284884639353077760x^{35}y^3 \\
& + 2804400012308152320x^{35}y^2 - 4455881124981424128x^{35}y + 655339962256441344x^{35} \\
& - 1479058834194432x^{34}y^8 + 23688155275001856x^{34}y^7 - 85841789065297920x^{34}y^6 \\
& - 80447038037213184x^{34}y^5 + 1091939227340783616x^{34}y^4 - 3425493212909051904x^{34}y^3 \\
& + 8547769933304979456x^{34}y^2 - 15926205610947981312x^{34}y + 13093011830099312640x^{34} \\
& + 1333142716022784x^{33}y^9 - 9935561137324032x^{33}y^8 - 52359014358024192x^{33}y^7 \\
& + 710798035987906560x^{33}y^6 - 2643195864482611200x^{33}y^5 + 3931842040617148416x^{33}y^4 \\
& + 930498380357640192x^{33}y^3 - 9974902734349996032x^{33}y^2 + 464725571017829904x^{33}y \\
& + 11433540669986469888x^{33} - 590297023512576x^{32}y^{10} - 6907801685262336x^{32}y^9 \\
& + 160666911208636416x^{32}y^8 - 871231978848780288x^{32}y^7 + 1940005364292108288x^{32}y^6 \\
& - 1178556195783868416x^{32}y^5 - 4202922683563646976x^{32}y^4 \\
& + 18901558998994231296x^{32}y^3 - 46589536851571372032x^{32}y^2 + 60693427360022851584x^{32}y \\
& - 25731332143697885184x^{32} - 368106570842112x^{31}y^{11} + 1667091650110784x^{31}y^{10} \\
& - 149728176758882304x^{31}y^9 + 410864967879524352x^{31}y^8 + 615104421739462656x^{31}y^7 \\
& - 6730664492215074816x^{31}y^6 + 18032817957515034624x^{31}y^5 \\
& - 21500344546749333504x^{31}y^4 + 7519580711831420928x^{31}y^3 \\
& - 6426454257892786176x^{31}y^2 + 46404203069306474496x^{31}y - 53690026159826915328x^{31} \\
& + 666571358011392x^{30}y^{12} - 14364446951374848x^{30}y^{11} + 67273963027955712x^{30}y^{10} \\
& + 210818115210461184x^{30}y^9 - 2711218476685049856x^{30}y^8 + 9744905562090135552x^{30}y^7 \\
& - 18735380023514775552x^{30}y^6 + 21370104255187685376x^{30}y^5 \\
& - 4619044155239976960x^{30}y^4 - 5131213098114677600x^{30}y^3 \\
& + 118439726670207461376x^{30}y^2 - 91769958070966103040x^{30}y - 5539550494144665600x^{30} \\
& - 457646006992896x^{29}y^{13} + 4279653420466176x^{29}y^{12} + 35625088624066560x^{29}y^{11} \\
& - 577715903664537600x^{29}y^{10} + 2724353414527057920x^{29}y^9 \\
& - 5448735431929552896x^{29}y^8 + 283594193682997248x^{29}y^7 \\
& + 22558319932180746240x^{29}y^6 - 52024206508603932672x^{29}y^5 \\
& + 58757801033779992576x^{29}y^4 - 64726042823385901056x^{29}y^3 \\
& + 134636109187235097600x^{29}y^2 - 188168520445569395712x^{29}y + 87393897078426828288x^{29} \\
& + 99488262389760x^{28}y^{14} + 3032733865181184x^{28}y^{13} - 75881355862376448x^{28}y^{12} \\
& + 534946728732033024x^{28}y^{11} - 1383366878083694592x^{28}y^{10} \\
& - 1114347300306370560x^{28}y^9 + 16504214383184928768x^{28}y^8 \\
& - 49876428198999343104x^{28}y^7 + 91950446101234176000x^{28}y^6 \\
& - 119580162176485380096x^{28}y^5 + 85260767145947449344x^{28}y^4 \\
& + 23363182047586037760x^{28}y^3 - 61515002726257572864x^{28}y^2 - 61691369144355330048x^{28}y \\
& + 108538941473642958336x^{28} + 19897652477952x^{27}y^{15} - 3758998180626432x^{27}y^{14} \\
& + 48556904597028864x^{27}y^{13} - 169872891755102208x^{27}y^{12} - 452202340902469632x^{27}y^{11} \\
& + 5419726727288832000x^{27}y^{10} - 18902043589738954752x^{27}y^9 \\
& + 34417653887451930624x^{27}y^8 - 32856519723992506368x^{27}y^7 \\
& + 9674070955604692992x^{27}y^6 + 11627227217314234368x^{27}y^5 \\
& - 35238769794091542528x^{27}y^4 + 149065528243783661568x^{27}y^3 \\
& - 302264086127778800640x^{27}y^2 + 220908716005969234944x^{27}y - 1677662977516835328x^{27} \\
& - 19897652477952x^{26}y^{16} + 1425998427586560x^{26}y^{15} - 6980759744348160x^{26}y^{14} \\
& - 113830324482097152x^{26}y^{13} + 1316384747292082176x^{26}y^{12} \\
& - 5678692601617244160x^{26}y^{11} + 11319839162077446144x^{26}y^{10} \\
& - 3002344968167018496x^{26}y^9 - 40885743361353191424x^{26}y^8 \\
& + 124032565668528248832x^{26}y^7 - 225436876130828869632x^{26}y^6 \\
& + 294242657395776353280x^{26}y^5 - 254179893895112088576x^{26}y^4 \\
& + 201061858426258401792x^{26}y^3 - 322944236407460915712x^{26}y^2 \\
& + 380350373132723551488x^{26}y - 155240015989324778496x^{26} - 142599842758656x^{25}y^{16} \\
& - 7499756846481408x^{25}y^{15} + 137273075376537600x^{25}y^{14} - 829798433838858240x^{25}y^{13} \\
& + 1937249124889485312x^{25}y^{12} + 1593396134280290304x^{25}y^{11} \\
& - 21867675730946445312x^{25}y^{10} + 66992019095825104896x^{25}y^9 \\
& - 130153149067185690624x^{25}y^8 + 199907159023535966208x^{25}y^7 \\
& - 277345408103103458304x^{25}y^6 + 341242095474502152192x^{25}y^5 \\
& - 268688386507600452096x^{25}y^4 + 29741829215934352896x^{25}y^3 \\
& + 34734791908990722816x^{25}y^2 + 162539636430746771712x^{25}y - 176347086780410202240x^{25} \\
& - 66325508259840x^{24}y^{17} + 4307841761476608x^{24}y^{16} - 42859543437508608x^{24}y^{15} \\
& + 50003629745946624x^{24}y^{14} + 1334825725794877440x^{24}y^{13} \\
& - 9196098460269502464x^{24}y^{12} + 29254588637613907968x^{24}y^{11}
\end{aligned}$$

$$\begin{aligned}
& -54018528526083796992x^{24}y^{10} + 60022139636572545024x^{24}y^9 \\
& -31620559526244993024x^{24}y^8 - 26668999896963987456x^{24}y^7 \\
& + 89263209529282194432x^{24}y^6 - 111774105738537891840x^{24}y^5 \\
& + 126146579086839845376x^{24}y^4 - 331609111383584715264x^{24}y^3 \\
& + 554037885142366189824x^{24}y^2 - 332517675594542287104x^{24}y + 1671462169255992960x^{24} \\
& - 620143502229504x^{23}y^{17} - 5385631270699008x^{23}y^{16} + 190806051230760960x^{23}y^{15} \\
& - 1543774290741264384x^{23}y^{14} + 6043359365787230208x^{23}y^{13} \\
& - 12145859114617626624x^{23}y^{12} + 7345499259310018560x^{23}y^{11} \\
& + 26796267251251494912x^{23}y^{10} - 104570326265398806528x^{23}y^9 \\
& + 239645539075639891968x^{23}y^8 - 432721542741243429888x^{23}y^7 \\
& + 643084244264415891456x^{23}y^6 - 764781397822716931584x^{23}y^5 \\
& + 639449136096561252864x^{23}y^4 - 513002996465566351104x^{23}y^3 \\
& + 662185721244029419008x^{23}y^2 - 598086780172915526784x^{23}y + 207325264121657299584x^{23} \\
& - 86223160737792x^{22}y^{18} + 6841476177002496x^{22}y^{17} - 85103917785907200x^{22}y^{16} \\
& + 369384165944967168x^{22}y^{15} + 68800278786785280x^{22}y^{14} \\
& - 7073779197010452480x^{22}y^{13} + 33237375301125439488x^{22}y^{12} \\
& - 88343022848237236224x^{22}y^{11} + 169040623317882310656x^{22}y^{10} \\
& - 276783336365982087168x^{22}y^9 + 428210249814935912448x^{22}y^8 \\
& - 609942836532978091008x^{22}y^7 + 787042208391856561152x^{22}y^6 \\
& - 898900428684046017024x^{22}y^5 + 690418637922268237824x^{22}y^4 \\
& - 230016240208765824768x^{22}y^3 + 112511437401922404480x^{22}y^2 \\
& - 279833146214127326208x^{22}y + 204324679094764986432x^{22} - 1051259305918464x^{21}y^{18} \\
& + 1359672919326720x^{21}y^{17} + 175085247635472384x^{21}y^{16} - 1886395285034532864x^{21}y^{15} \\
& + 9539591634996387840x^{21}y^{14} - 29499591748850343936x^{21}y^{13} \\
& + 64073682845488852992x^{21}y^{12} - 109556765132236775424x^{21}y^{11} \\
& + 15805985310822998592x^{21}y^{10} - 192827351252359120896x^{21}y^9 \\
& + 189296623138120823808x^{21}y^8 - 129026784621118703616x^{21}y^7 \\
& + 54872140848670932480x^{21}y^6 - 49378173907314728448x^{21}y^5 \\
& + 1788181129407714048x^{21}y^4 + 343333552278487362048x^{21}y^3 \\
& - 603644643823849785216x^{21}y^2 + 324460228158477286848x^{21}y - 11827147833938346624x^{21} \\
& - 66325508259840x^{20}y^{19} + 7756768190988288x^{20}y^{18} - 118915003759067136x^{20}y^{17} \\
& + 794391390323195904x^{20}y^{16} - 2712522601991208960x^{20}y^{15} \\
& + 4378957215036088320x^{20}y^{14} + 464977048327741440x^{20}y^{13} \\
& - 19251631456873279488x^{20}y^{12} + 57843528256315662336x^{20}y^{11} \\
& - 131886178014066075648x^{20}y^{10} + 280458619111884589056x^{20}y^9 \\
& - 537514214184287437824x^{20}y^8 + 871320717922586867712x^{20}y^7 \\
& - 1161670270457669617152x^{20}y^6 + 1249364460385530441216x^{20}y^5 \\
& - 961505893938925373184x^{20}y^4 + 751640321164492196352x^{20}y^3 \\
& - 82033955734329746176x^{20}y^2 + 606180710475872233920x^{20}y - 204151136868389929536x^{20} \\
& - 1329826440609792x^{19}y^{19} + 13420966596378624x^{19}y^{18} + 22387346244255744x^{19}y^{17} \\
& - 1017488015250259968x^{19}y^{16} + 7121382672426246144x^{19}y^{15} \\
& - 27670441180926025728x^{19}y^{14} + 75026125534196994048x^{19}y^{13} \\
& - 163838034490405036032x^{19}y^{12} + 314496781033199966208x^{19}y^{11} \\
& - 535800375435069167616x^{19}y^{10} + 819297993530008525824x^{19}y^9 \\
& - 1149660046463366928384x^{19}y^8 + 1463137878221554383360x^{19}y^7 \\
& - 1663810975468577401344x^{19}y^6 + 1713036197705778286848x^{19}y^5 \\
& - 1265866883135091730944x^{19}y^4 + 625922660916787206528x^{19}y^3 \\
& - 388058184495049953600x^{19}y^2 + 334717865627366898432x^{19}y - 181207377252613185696x^{19} \\
& - 33162754129920x^{18}y^{20} + 5501700910153728x^{18}y^{19} - 103658478721597440x^{18}y^{18} \\
& + 881972565842608128x^{18}y^{17} - 4444847876682399744x^{18}y^{16} \\
& + 15675410002901114880x^{18}y^{15} - 43608200902680084480x^{18}y^{14} \\
& + 103982671380687925248x^{18}y^{13} - 218889455131032440832x^{18}y^{12} \\
& + 398827694088922355712x^{18}y^{11} - 608226129930962829312x^{18}y^{10} \\
& + 780989985554592863232x^{18}y^9 - 872913027924852599808x^{18}y^8 \\
& + 853665398927410383360x^{18}y^7 - 741983808197394911232x^{18}y^6 \\
& + 708883150710820209408x^{18}y^5 - 486497912302683097344x^{18}y^4 \\
& - 529099080016201344x^{18}y^3 + 286052401309769557056x^{18}y^2 - 179059317553261480608x^{18}y \\
& - 4526217752210111232x^{18} - 1316561338957824x^{17}y^{20} + 22394807863934976x^{17}y^{19} \\
& - 165824548544692224x^{17}y^{18} + 661300138242441216x^{17}y^{17} \\
& - 1675049467498979328x^{17}y^{16} + 3613453070766612480x^{17}y^{15} \\
& - 9342697523909013504x^{17}y^{14} + 25207603027451117568x^{17}y^{13} \\
& - 52993845540424230912x^{17}y^{12} + 72856804181341575168x^{17}y^{11} \\
& - 43614951958535279616x^{17}y^{10} - 76631079095558774784x^{17}y^9 \\
& + 309887834599783328256x^{17}y^8 - 623346527857426085376x^{17}y^7 \\
& + 914969305698704835840x^{17}y^6 - 946349815349319238656x^{17}y^5 \\
& + 729100954888521471360x^{17}y^4 - 631693184182178361984x^{17}y^3 \\
& + 638578552473591491808x^{17}y^2 - 468004605601352108832x^{17}y + 108071044645808527632x^{17} \\
& + 13265101651968x^{16}y^{21} + 971668696006656x^{16}y^{20} - 30891105472020480x^{16}y^{19} \\
& + 326141592697257984x^{16}y^{18} - 1881520360177434624x^{16}y^{17} \\
& + 7548284319134146560x^{16}y^{16} - 2436399106291386776x^{16}y^{15} \\
& + 66905049151317749760x^{16}y^{14} - 156020904283418320896x^{16}y^{13} \\
& + 310242699619710584832x^{16}y^{12} - 543709651048879638528x^{16}y^{11}
\end{aligned}$$

$$\begin{aligned}
& +840407554503740888064x^{16}y^{10} - 1163542069234199070720x^{16}y^9 \\
& +1498451851659537575424x^{16}y^8 - 1777341031272416042496x^{16}y^7 \\
& +1917213059009002998528x^{16}y^6 - 1833999334710987373056x^{16}y^5 \\
& +1319216544965732582016x^{16}y^4 - 764415409569809127552x^{16}y^3 \\
& +425392362090112023648x^{16}y^2 - 363555800073785744352x^{16}y + 44239707733910368752x^{16} \\
& -732896866271232x^{15}y^{21} + 14976299765071872x^{15}y^{20} - 143081531762393088x^{15}y^{19} \\
& +904907097530007552x^{15}y^{18} - 4259703951184896000x^{15}y^{17} \\
& +15989112246524239872x^{15}y^{16} - 50462166794689548288x^{15}y^{15} \\
& +134512926085202706432x^{15}y^{14} - 300322258418330449920x^{15}y^{13} \\
& +563278478238256416768x^{15}y^{12} - 908107907953822866432x^{15}y^{11} \\
& +1260036735301232345088x^{15}y^{10} - 1530785837214584861184x^{15}y^9 \\
& +1690956630978780704256x^{15}y^8 - 1682071708512987263232x^{15}y^7 \\
& +1503931936055032719360x^{15}y^6 - 1317459536370140805504x^{15}y^5 \\
& +872664138840035373696x^{15}y^4 - 394262716379398974144x^{15}y^3 \\
& -89113788423026057664x^{15}y^2 - 91971056356127038224x^{15}y - 73556720825116471344x^{15} \\
& +33162754129920x^{14}y^{22} - 1190542873264128x^{14}y^{21} + 14145572774117376x^{14}y^{20} \\
& -105443463962640384x^{14}y^{19} + 702087838615683072x^{14}y^{18} \\
& -3795342486480691200x^{14}y^{17} + 15090544623980593152x^{14}y^{16} \\
& -45102796694451597312x^{14}y^{15} + 107705435936152006656x^{14}y^{14} \\
& -218563142646517807104x^{14}y^{13} + 385516668758668849152x^{14}y^{12} \\
& -589462250199266423808x^{14}y^{11} + 75432550989994656768x^{14}y^{10} \\
& -80037757293796725248x^{14}y^9 + 709458041380529304576x^{14}y^8 \\
& -467649680226102067968x^{14}y^7 + 139490638515653425920x^{14}y^6 \\
& -6541056786140005248x^{14}y^5 - 104130038425065889536x^{14}y^4 \\
& +68282463414076321536x^{14}y^3 - 456104135414870298288x^{14}y^2 \\
& +72968362829081470512x^{14}y - 34760276707234701528x^{14} - 142599842758656x^{13}y^{22} \\
& +2931587465084928x^{13}y^{21} - 21821921286340608x^{13}y^{20} + 116523969186299904x^{13}y^{19} \\
& -481970472612765696x^{13}y^{18} + 1469309399014662144x^{13}y^{17} \\
& -3494048294582489088x^{13}y^{16} + 9015397514756554752x^{13}y^{15} \\
& -27317913746538903552x^{13}y^{14} + 68579634858426943488x^{13}y^{13} \\
& -129530444726718913536x^{13}y^{12} + 204518235723462623232x^{13}y^{11} \\
& -310101018773867352576x^{13}y^{10} + 458279760204428004864x^{13}y^9 \\
& -642627337514801875200x^{13}y^8 + 843928885376288234496x^{13}y^7 \\
& -1009929420890668563840x^{13}y^6 + 945831474749555224704x^{13}y^5 \\
& -791259632332540136064x^{13}y^4 + 255707001564583580064x^{13}y^3 \\
& -464774333617515560544x^{13}y^2 + 144152917168431567240x^{13}y + 215400182283371849832x^{13} \\
& +13265101651968x^{12}y^{23} - 666571358011392x^{12}y^{22} + 8580862631116800x^{12}y^{21} \\
& -73660280404525056x^{12}y^{20} + 436756788166459392x^{12}y^{19} \\
& -2168766602158989312x^{12}y^{18} + 9112922956636250112x^{12}y^{17} \\
& -29704190811325378560x^{12}y^{16} + 76994305767130669056x^{12}y^{15} \\
& -169340186310381324288x^{12}y^{14} + 326570784957573801984x^{12}y^{13} \\
& -553236607311834903552x^{12}y^{12} + 827858924775908407296x^{12}y^{11} \\
& -1088273271127220522496x^{12}y^{10} + 1262505191251165263360x^{12}y^9 \\
& -1339352139240067891968x^{12}y^8 + 1307821988783205222912x^{12}y^7 \\
& -1180980834642387119232x^{12}y^6 + 935340362107845910848x^{12}y^5 \\
& -808115046631467457824x^{12}y^4 + 136048356195467580000x^{12}y^3 \\
& -173266258190318260560x^{12}y^2 + 304838666827319245608x^{12}y + 505874648244860904312x^{12} \\
& +9948826238976x^{11}y^{23} - 840675817193472x^{11}y^{22} + 12499871100420096x^{11}y^{21} \\
& -117497296020013056x^{11}y^{20} + 690248320856875008x^{11}y^{19} \\
& -2993468749756932096x^{11}y^{18} + 11518174194831765504x^{11}y^{17} \\
& -37258058287384510464x^{11}y^{16} + 96327388368056051712x^{11}y^{15} \\
& -202153103437976598528x^{11}y^{14} + 362873430982842946560x^{11}y^{13} \\
& -580785404453420986368x^{11}y^{12} + 836000523229654107648x^{11}y^{11} \\
& -1043261802740421001728x^{11}y^{10} + 1113853881027650280192x^{11}y^9 \\
& -1050780323565448750080x^{11}y^8 + 857379161683306502784x^{11}y^7 \\
& -594106767300878564544x^{11}y^6 + 305494338182346374688x^{11}y^5 \\
& -410813967518581756032x^{11}y^4 - 29597423787750792672x^{11}y^3 \\
& +280103322144852662904x^{11}y^2 + 569087020923665077008x^{11}y + 581147974885941341532x^{11} \\
& -109437088628736x^{10}y^{23} - 296806649462784x^{10}y^{22} + 6466737055334400x^{10}y^{21} \\
& -82447581180100608x^{10}y^{20} + 575860447570968576x^{10}y^{19} \\
& -2220954413798817792x^{10}y^{18} + 6452757710865358848x^{10}y^{17} \\
& -17812019096914747392x^{10}y^{16} + 46547044585635852288x^{10}y^{15} \\
& -100459613137861435392x^{10}y^{14} + 174003652182045542400x^{10}y^{13} \\
& -255427150104168302592x^{10}y^{12} + 336186331372842882048x^{10}y^{11} \\
& -380003449284036126720x^{10}y^{10} + 353748392195344141824x^{10}y^9 \\
& -270741953318982188544x^{10}y^8 + 123435026501312293632x^{10}y^7 \\
& +34010437802677720128x^{10}y^6 - 248546975452187927904x^{10}y^5 \\
& -43721855897304793680x^{10}y^4 + 26392379743779240384x^{10}y^3 \\
& +723233014286505236352x^{10}y^2 + 712035777249888983244x^{10}y + 312142110921467087472x^{10} \\
& -316704301940736x^9y^{23} + 3367677681893376x^9y^{22} - 17674089813540864x^9y^{21} \\
& +72989563702247424x^9y^{20} - 130332940005998592x^9y^{19} + 102206779159560192x^9y^{18} \\
& -1298678323793264640x^9y^{17} + 8142538488283275264x^9y^{16}
\end{aligned}$$

$$\begin{aligned}
& -24080746321752514560x^9y^{15} + 48431763907983544320x^9y^{14} \\
& -83584022336277184512x^9y^{13} + 136017869796232427520x^9y^{12} \\
& -201007312665074731008x^9y^{11} + 265280787165476256768x^9y^{10} \\
& -306800858930786795520x^9y^9 + 314866857445728993792x^9y^8 \\
& -316393693128516928320x^9y^7 + 280535141299791681216x^9y^6 \\
& -401361299035138105200x^9y^5 + 130810249821138425856x^9y^4 \\
& + 299875077634150343376x^9y^3 + 932503215309583013388x^9y^2 + 530661390828365115972x^9y \\
& -169171666220558869614x^9 - 441064629927936x^8y^{23} + 7839675076313088x^8y^{22} \\
& -41814916682416128x^8y^{21} + 217355323118321664x^8y^{20} - 1006429894885638144x^8y^{19} \\
& + 3163012915711721472x^8y^{18} - 8406942225220141056x^8y^{17} \\
& + 23346093487650103296x^8y^{16} - 60109220054504595456x^8y^{15} \\
& + 125652970897008832512x^8y^{14} - 213118650699879997440x^8y^{13} \\
& + 309834093035387308032x^8y^{12} - 402782011978883844096x^8y^{11} \\
& + 468248773674087355392x^8y^{10} - 476580725326111131648x^8y^9 \\
& + 428625620901574848000x^8y^8 - 350075585785224475584x^8y^7 \\
& + 208936197031412970432x^8y^6 - 272987609428497714624x^8y^5 \\
& + 208293287577686096904x^8y^4 + 538552731051503199504x^8y^3 \\
& + 766315612216399503348x^8y^2 + 96384420445651359156x^8y - 543722995751761463382x^8 \\
& + 338260092125184x^7y^{23} + 4224934876151808x^7y^{22} - 41458417075519488x^7y^{21} \\
& + 192690524734193664x^7y^{20} - 1030094836232749056x^7y^{19} + 4125187944279834624x^7y^{18} \\
& - 11404637982525358080x^7y^{17} + 25313424546810839040x^7y^{16} \\
& - 51978798407511539712x^7y^{15} + 98157437335716470784x^7y^{14} \\
& - 159699117165915414528x^7y^{13} + 222407006905802870784x^7y^{12} \\
& - 273757971161134964736x^7y^{11} + 298708567922125928448x^7y^{10} \\
& - 285940056128394442752x^7y^9 + 244651131350209612800x^7y^8 \\
& - 184318990536384067680x^7y^7 + 57751035856362383040x^7y^6 \\
& - 88263961235644573368x^7y^5 + 258825219348811577688x^7y^4 \\
& + 531303434346245345712x^7y^3 + 336471967095366413592x^7y^2 - 291600672907448312766x^7y \\
& - 583306007996107870506x^7 + 1273449758588928x^6y^{23} - 5332570864091136x^6y^{22} \\
& - 15546699136106496x^6y^{21} + 82764285482041344x^6y^{20} - 329625340018606080x^6y^{19} \\
& + 1799134130092474368x^6y^{18} - 6355394351946473472x^6y^{17} \\
& + 13756573253638987776x^6y^{16} - 21651258084701147136x^6y^{15} \\
& + 30215693338292477952x^6y^{14} - 41362747607901394944x^6y^{13} \\
& + 54202356615677589504x^6y^{12} - 64688192876677340160x^6y^{11} \\
& + 65699402405318756352x^6y^{10} - 57608015055799527936x^6y^9 \\
& + 48034507353986437632x^6y^8 - 34845715191666919680x^6y^7 \\
& - 28681897352770979424x^6y^6 + 34342312296909217320x^6y^5 \\
& + 248400309104851911444x^6y^4 + 312116266448113487796x^6y^3 \\
& - 57584121535847796084x^6y^2 - 402377121482380246530x^6y - 354858573627781979958x^6 \\
& - 984933797658624x^5y^{23} - 4612939099471872x^5y^{22} + 14972983489658880x^5y^{21} \\
& + 5422939369095168x^5y^{20} + 91739784887304192x^5y^{19} - 606948456895635456x^5y^{18} \\
& + 1171879289774235648x^5y^{17} - 1811803338506170368x^5y^{16} \\
& + 4966325138290139136x^5y^{15} - 12954645938340587520x^5y^{14} \\
& + 24051405885936887808x^5y^{13} - 33876135285370610688x^5y^{12} \\
& + 3997787410444135424x^5y^{11} - 43018746860832450048x^5y^{10} \\
& + 41072859667560973824x^5y^9 - 33436947369379809024x^5y^8 \\
& + 23329788246417134736x^5y^7 - 38174853214208379888x^5y^6 \\
& + 74631037613853499668x^5y^5 + 162408111342450591852x^5y^4 \\
& + 76501120009115312946x^5y^3 - 207911468973792630312x^5y^2 - 274883952260690709984x^5y \\
& - 103705526746919161944x^5 - 3011178074996736x^4y^{23} + 6347351140466688x^4y^{22} \\
& + 22320191667142656x^4y^{21} - 76792502532096000x^4y^{20} + 209983242875240448x^4y^{19} \\
& - 1025776631110606848x^4y^{18} + 3392763232720527360x^4y^{17} \\
& - 7481928342593949696x^4y^{16} + 12875985232781303808x^4y^{15} \\
& - 19454187382793091072x^4y^{14} + 27375692798525036544x^4y^{13} \\
& - 35615917930763160576x^4y^{12} + 41583150795391469568x^4y^{11} \\
& - 42918269985651995136x^4y^{10} + 38867407338630597120x^4y^9 \\
& - 31708327229302185216x^4y^8 + 21849816199406276592x^4y^7 \\
& - 16842420963439313328x^4y^6 + 58298888066283662148x^4y^5 \\
& + 60775216500923309376x^4y^4 - 37350239761059032466x^4y^3 \\
& - 155668371282348152616x^4y^2 - 105425420939558924058x^4y + 16754153489351758584x^4 \\
& - 56376682020864x^3y^{23} + 4017667662839808x^3y^{22} + 13282512097886208x^3y^{21} \\
& - 113038563727245312x^3y^{20} + 314312023764688896x^3y^{19} - 753907543684669440x^3y^{18} \\
& + 1918046024843341824x^3y^{17} - 4200540768748265472x^3y^{16} \\
& + 7135869717934295040x^3y^{15} - 9787831327402297344x^3y^{14} \\
& + 12057123408903263232x^3y^{13} - 14353005697566418944x^3y^{12} \\
& + 16142910866360151552x^3y^{11} - 16215103487627214336x^3y^{10} \\
& + 14466636333042690816x^3y^9 - 12049234481955386880x^3y^8 + 8196091912825790472x^3y^7 \\
& - 1272325097116868076x^3y^6 + 25870145272004442132x^3y^5 + 3910985649575178966x^3y^4 \\
& - 41983215298758006156x^3y^3 - 60888516717004677702x^3y^2 - 15824214977647638492x^3y \\
& + 29160310309600604418x^3 + 2205323149639680x^2y^{23} - 6926041200033792x^2y^{22} \\
& + 16783669865152512x^2y^{21} - 72702705879023616x^2y^{20} + 233347232228622336x^2y^{19} \\
& - 472520331289018368x^2y^{18} + 704235541480022016x^2y^{17} - 993062817764720640x^2y^{16}
\end{aligned}$$

$$\begin{aligned}
& +1496761319616196608x^2y^{15} - 2116118035503968256x^2y^{14} \\
& +2574798202257684480x^2y^{13} - 2746946162579705856x^2y^{12} \\
& +2682779705064133632x^2y^{11} - 2477764864559195136x^2y^{10} + 2270524310992507392x^2y^9 \\
& -2044770079947955200x^2y^8 + 1383894826578853056x^2y^7 + 1947795275160405684x^2y^6 \\
& +5460976845911500704x^2y^5 - 7109273677572209178x^2y^4 - 16279289521882474914x^2y^3 \\
& -11456007642473254938x^2y^2 + 4238932192748822628x^2y + 12166044601210245126x^2 \\
& +1009805863256064xy^{23} - 6014065461460992xy^{22} + 16178449602281472xy^{21} \\
& -32404985198051328xy^{20} + 69620227882647552xy^{19} - 142551756765167616xy^{18} \\
& +201184335135719424xy^{17} - 157644126169694208xy^{16} + 44964134721478656xy^{15} \\
& -4825595260329984xy^{14} + 48005781076832256xy^{13} - 25964363811594240xy^{12} \\
& -85122260932485312xy^{11} + 138782498857482240xy^{10} - 81555762178022400xy^9 \\
& +3257401610507264xy^8 - 19976668127808264xy^7 + 725898891340403712xy^6 \\
& -104972593286987136xy^5 - 2768602949500700988xy^4 - 2615734573573660488xy^3 \\
& -27479200056121980xy^2 + 2363907530658524256xy + 2344982787530383968x + 69641783672832y^{23} \\
& -998198899310592y^{22} + 4387432371388416y^{21} - 9600617320611840y^{20} + 11006718095720448y^{19} \\
& -6370565068357632y^{18} + 6687269370298368y^{17} - 19022155768922112y^{16} \\
& +22330969562234880y^{15} + 8276594361974784y^{14} - 57380269867720704y^{13} \\
& +89443264163807232y^{12} - 93758360277749760y^{11} + 78571891558379520y^{10} \\
& -57574168345774080y^9 + 48279995600044032y^8 - 34806364592538744y^7 + 63686729214791352y^6 \\
& -158027337320854464y^5 - 272870046282107988y^4 - 51242271511402872y^3 \\
& +248798635304311188y^2 + 307535522214856644y + 133303438883759616, \\
& 291832236343296x^{42} + 106120813215744x^{41}y - 99488262389760x^{41} - 504073862774784x^{40}y^2 \\
& +7196317646192640x^{40}y - 21416506617102336x^{40} + 845650230312960x^{39}y^3 \\
& -3220103426015232x^{39}y^2 + 7310729147940864x^{39}y - 45207466429906944x^{39} \\
& -703050387554304x^{38}y^4 - 1954944355958784x^{38}y^3 + 56726549076934656x^{38}y^2 \\
& -230110547425542144x^{38}y + 284685662828298240x^{38} - 298464787169280x^{37}y^5 \\
& +12825695159746560x^{37}y^4 - 65940820311932928x^{37}y^3 + 159901680657088512x^{37}y^2 \\
& -462203398479200256x^{37}y + 921089277942128640x^{37} + 1369621745565696x^{36}y^6 \\
& -20277366012739584x^{36}y^5 + 67580718503657472x^{36}y^4 + 84647929916620800x^{36}y^3 \\
& -1000225972656783360x^{36}y^2 + 2148415863552737280x^{36}y - 1055621451689828352x^{36} \\
& -6634214974439682048x^{35}y^7 + 11403013007572992x^{35}y^6 + 28521626689437696x^{35}y^5 \\
& -354213034999382016x^{35}y^4 + 1132083570283905024x^{35}y^3 - 2794494297649545216x^{35}y^2 \\
& +6120430849988444160x^{35}y - 6625381038541111296x^{35} + 669887633424384x^{34}y^8 \\
& +7488149882535936x^{34}y^7 - 148680233728376832x^{34}y^6 + 718760413254500352x^{34}y^5 \\
& -1279069187276242944x^{34}y^4 - 671807342354079744x^{34}y^3 + 6136808276115505152x^{34}y^2 \\
& -6634214974439682048x^{34}y - 2492148846445424640x^{34} + 404585600385024x^{33}y^9 \\
& -21887417725747200x^{33}y^8 + 175485687891591168x^{33}y^7 - 462264749574340608x^{33}y^6 \\
& +11990822824525824x^{33}y^5 + 2529048421164146688x^{33}y^4 - 8716345746839175168x^{33}y^3 \\
& +21804434358820687872x^{33}y^2 - 34220923310195859456x^{33}y + 20950973367207671808x^{33} \\
& -1101003437113344x^{32}y^{10} + 20139740583100416x^{32}y^9 - 77076873148760064x^{32}y^8 \\
& -224898191545171968x^{32}y^7 + 2488242066741706752x^{32}y^6 - 7673150793656254464x^{32}y^5 \\
& +10021246646910492672x^{32}y^4 - 460351258661044224x^{32}y^3 \\
& -9402534324788908032x^{32}y^2 - 9640605384076124160x^{32}y + 28810717505984120832x^{32} \\
& +892078086094848x^{31}y^{11} - 8814660047732736x^{31}y^{10} - 49052687771271168x^{31}y^9 \\
& +769720788457095168x^{31}y^8 - 3261246796926517248x^{31}y^7 + 6620141323388583936x^{31}y^6 \\
& -6240306744548425728x^{31}y^5 - 3742515462755008512x^{31}y^4 \\
& +34542309778485313536x^{31}y^3 - 82762995036371263488x^{31}y^2 + 86658504170830884864x^{31}y \\
& -19613201441609170944x^{31} - 271934583865344x^{30}y^{12} - 5131936201605120x^{30}y^{11} \\
& +122186509453983744x^{30}y^{10} - 741919622601129984x^{30}y^9 + 1669895560972763136x^{30}y^8 \\
& +977875519976497152x^{30}y^7 - 14202430730607550464x^{30}y^6 \\
& +34443501559822430208x^{30}y^5 - 39068652221701005312x^{30}y^4 \\
& +27018377595144013824x^{30}y^3 - 52037078631619461120x^{30}y^2 \\
& +110134488031169762304x^{30}y - 78270542388852986880x^{30} - 212241626431488x^{29}y^{13} \\
& +10159409727700992x^{29}y^{12} - 108220015552167936x^{29}y^{11} + 350579226215596032x^{29}y^{10} \\
& +597456033060372480x^{29}y^9 - 7441744826147512320x^{29}y^8 \\
& +24049284402375032832x^{29}y^7 - 44929138274312564736x^{29}y^6 \\
& +56043493964715773952x^{29}y^5 - 30741687470646319104x^{29}y^4 \\
& -55402498759025442816x^{29}y^3 + 124571274763533124608x^{29}y^2 \\
& -48493723643295787008x^{29}y - 51789532014866465280x^{29} + 235455554322432x^{28}y^{14} \\
& -6627576412864512x^{28}y^{13} + 33608793172967424x^{28}y^{12} + 151640009534472192x^{28}y^{11} \\
& -2003080922647216128x^{28}y^{10} + 7916198131467878400x^{28}y^9 \\
& -14747328772785119232x^{28}y^8 + 7985879295911239680x^{28}y^7 \\
& +21783838215833874432x^{28}y^6 - 54657563398470623232x^{28}y^5 \\
& +66806062164683384832x^{28}y^4 - 115083343393301882880x^{28}y^3 \\
& +236367561099269993472x^{28}y^2 - 243072130027527447552x^{28}y + 64267494763572269568x^{28} \\
& -99488262389760x^{27}y^{15} + 1162354532253696x^{27}y^{14} + 22933702618546176x^{27}y^{13} \\
& -367352118185656320x^{27}y^{12} + 1953583024901750784x^{27}y^{11} \\
& -4333110951054753792x^{27}y^{10} - 868495222564208640x^{27}y^9 \\
& +29609647322843602944x^{27}y^8 - 87289257059908190208x^{27}y^7 \\
& +159334390086986354688x^{27}y^6 - 209666413591170084864x^{27}y^5 \\
& +169014903218481604608x^{27}y^4 - 65033152829656031232x^{27}y^3 \\
& +96227767713733788672x^{27}y^2 - 224630244989757960192x^{27}y + 152121791005294818816x^{27}
\end{aligned}$$

$$\begin{aligned}
& +13265101651968x^{26}y^{16} + 759427069575168x^{26}y^{15} - 27845106505187328x^{26}y^{14} \\
& + 232708849211621376x^{26}y^{13} - 645254339656679424x^{26}y^{12} \\
& - 999786151630135296x^{26}y^{11} + 11980609939857088512x^{26}y^{10} \\
& - 38353352134572748800x^{26}y^9 + 71026530121487499264x^{26}y^8 \\
& - 91899640036471891968x^{26}y^7 + 102917409412738922496x^{26}y^6 \\
& - 115013392418763592704x^{26}y^5 + 71713558546185572352x^{26}y^4 \\
& + 115769070257625561600x^{26}y^3 - 253065486383680250880x^{26}y^2 \\
& + 81702167174779246848x^{26}y + 88694622489806569728x^{26} - 494125036535808x^{25}y^{16} \\
& + 9864261215944704x^{25}y^{15} - 24710397171056640x^{25}y^{14} - 431839580797845504x^{25}y^{13} \\
& + 3786705247167553536x^{25}y^{12} - 13785055499557896192x^{25}y^{11} \\
& + 25867240260841156608x^{25}y^{10} - 18227147344104886272x^{25}y^9 \\
& - 32196696468278519808x^{25}y^8 + 124791823089889320960x^{25}y^7 \\
& - 231997008673367313408x^{25}y^6 + 296566762276356940800x^{25}y^5 \\
& - 270350935948548601344x^{25}y^4 + 324471661589958137856x^{25}y^3 \\
& - 521218182941184335616x^{25}y^2 + 439858874548008829440x^{25}y - 102299841232506612864x^{25} \\
& + 72958059085824x^{24}y^{17} + 102804537802752x^{24}y^{16} - 44005316592697344x^{24}y^{15} \\
& + 505102737221664768x^{24}y^{14} - 2399690880663994368x^{24}y^{13} \\
& + 5015265072697368576x^{24}y^{12} + 738941607280263168x^{24}y^{11} \\
& - 31573274309635239936x^{24}y^{10} + 98165901921440919552x^{24}y^9 \\
& - 202233437308115343360x^{24}y^8 + 341391038523057008640x^{24}y^7 \\
& - 499984488687742755840x^{24}y^6 + 611723459628249670656x^{24}y^5 \\
& - 504199838517523352064x^{24}y^4 + 275070693652218233856x^{24}y^3 \\
& - 318439962546469606656x^{24}y^2 + 430607949612647035392x^{24}y - 224047214122326408576x^{24} \\
& - 951771043528704x^{23}y^{17} + 22315217254023168x^{23}y^{16} - 138725604007428096x^{23}y^{15} \\
& + 29091196991619072x^{23}y^{14} + 3473545214574452736x^{23}y^{13} \\
& - 19190677278446419968x^{23}y^{12} + 54923935137597886464x^{23}y^{11} \\
& - 103027914031149821952x^{23}y^{10} + 147937695102243145728x^{23}y^9 \\
& - 188734434916825571328x^{23}y^8 + 227100135669249825792x^{23}y^7 \\
& - 269324854971443647488x^{23}y^6 + 319002054582823681536x^{23}y^5 \\
& - 223264964559608690688x^{23}y^4 - 138459752781044463360x^{23}y^3 \\
& + 329285239695160231680x^{23}y^2 - 60628756804019623296x^{23}y - 114232939627853083392x^{23} \\
& + 172446321475584x^{22}y^{18} - 1787472447602688x^{22}y^{17} - 41012378032472064x^{22}y^{16} \\
& + 661397139298271232x^{22}y^{15} - 4089567830068887552x^{22}y^{14} \\
& + 13944757789155778560x^{22}y^{13} - 28991983565554163712x^{22}y^{12} \\
& + 37349388719525523456x^{22}y^{11} - 23422040754486312960x^{22}y^{10} \\
& - 28420402045968414720x^{22}y^9 + 140890008774716540928x^{22}y^8 \\
& - 319450506967577705472x^{22}y^7 + 505381143931532992512x^{22}y^6 \\
& - 583322031677965091328x^{22}y^5 + 497985369030167956992x^{22}y^4 \\
& - 575083670353656664320x^{22}y^3 + 799464158410447476864x^{22}y^2 \\
& - 565133533899448853376x^{22}y + 128168049455863174080x^{22} - 1124217365004288x^{21}y^{18} \\
& + 31527830351314944x^{21}y^{17} - 275528597344174080x^{21}y^{16} + 984208362412032000x^{21}y^{15} \\
& - 401622922837942272x^{21}y^{14} - 9446920391252459520x^{21}y^{13} \\
& + 41875136305839009792x^{21}y^{12} - 106786873245540151296x^{21}y^{11} \\
& + 210112620002492258304x^{21}y^{10} - 373566690011198287872x^{21}y^9 \\
& + 623306730029499122688x^{21}y^8 - 930139170680112694272x^{21}y^7 \\
& + 1207223908317417060864x^{21}y^6 - 1332587373941178049536x^{21}y^5 \\
& + 1030887453613626240768x^{21}y^4 - 605540148573307647744x^{21}y^3 \\
& + 578777247357947884800x^{21}y^2 - 540777198815769570624x^{21}y + 238397178803888938176x^{21} \\
& + 245404380561408x^{20}y^{19} - 4122130338349056x^{20}y^{18} - 10991794856361984x^{20}y^{17} \\
& + 527965139918831616x^{20}y^{16} - 4272033448635973632x^{20}y^{15} \\
& + 18544566096129908736x^{20}y^{14} - 53712299301836636160x^{20}y^{13} \\
& + 119169801282468630528x^{20}y^{12} - 223513849474849640448x^{20}y^{11} \\
& + 367634922173088565248x^{20}y^{10} - 539092543287580028928x^{20}y^9 \\
& + 713243872478224714752x^{20}y^8 - 848003387061578480640x^{20}y^7 \\
& + 923657064113201167872x^{20}y^6 - 970189134755528696832x^{20}y^5 \\
& + 710930177609849074944x^{20}y^4 - 134609695032601868544x^{20}y^3 \\
& - 129990964380662279808x^{20}y^2 - 24831965110239412416x^{20}y + 105365880463073798016x^{20} \\
& - 1024729102614528x^{19}y^{19} + 32008690286198784x^{19}y^{18} - 358728973043023872x^{19}y^{17} \\
& + 2069704895694225408x^{19}y^{16} - 7231948952833105920x^{19}y^{15} \\
& + 17620935722152574976x^{19}y^{14} - 35493485757844598784x^{19}y^{13} \\
& + 68082282153211834368x^{19}y^{12} - 118155729756328998912x^{19}y^{11} \\
& + 156825271651985436672x^{19}y^{10} - 13155899722584634368x^{19}y^9 \\
& + 3159582591514604544x^{19}y^8 + 217318004371572420096x^{19}y^7 \\
& - 438365389120047922176x^{19}y^6 + 478400965121054041344x^{19}y^5 \\
& - 381192913120381555968x^{19}y^4 + 523249089547639026816x^{19}y^3 \\
& - 696591176550781780800x^{19}y^2 + 459813544174796142528x^{19}y - 109663913837683776096x^{19} \\
& + 252036931387392x^{18}y^{20} - 6556276491485184x^{18}y^{19} + 48828839180894208x^{18}y^{18} \\
& - 65896879662710784x^{18}y^{17} - 926603829419507712x^{18}y^{16} \\
& + 6581572625801060352x^{18}y^{15} - 23879828117718687744x^{18}y^{14} \\
& + 61192689721707810816x^{18}y^{13} - 129539312188089237504x^{18}y^{12} \\
& + 249556390644861720576x^{18}y^{11} - 442765365088478275584x^{18}y^{10} \\
& + 719892653570614324224x^{18}y^9 - 1084581979139487006720x^{18}y^8
\end{aligned}$$

$$\begin{aligned}
& +1466016889248974522880x^{18}y^7 - 1735886832065127876096x^{18}y^6 \\
& +1758590499325974177024x^{18}y^5 - 1301641452053412627456x^{18}y^4 \\
& +839982140989460484480x^{18}y^3 - 681926233300286336640x^{18}y^2 \\
& +495675670793853505248x^{18}y - 180036915606559265760x^{18} - 487492485709824x^{17}y^{20} \\
& +16282912277790720x^{17}y^{19} - 223230934081290240x^{17}y^{18} \\
& +1607152459227807744x^{17}y^{17} - 7409950035625451520x^{17}y^{16} \\
& +25312679213911769088x^{17}y^{15} - 70408254718651576320x^{17}y^{14} \\
& +167839560389939318784x^{17}y^{13} - 348114989941133727744x^{17}y^{12} \\
& +624851493243931901952x^{17}y^{11} - 957175796391617516544x^{17}y^{10} \\
& +1278686487901797190656x^{17}y^9 - 1560169668255761926656x^{17}y^8 \\
& +1733054384104415870976x^{17}y^7 - 1744709137148024163072x^{17}y^6 \\
& +1674426658918420852992x^{17}y^5 - 1183705395038350068096x^{17}y^4 \\
& +532786836108485440512x^{17}y^3 - 161146181819190034656x^{17}y^2 \\
& +132039732286587064320x^{17}y - 70931127279014230320x^{17} + 192343973953536x^{16}y^{21} \\
& -5806798248148992x^{16}y^{20} + 68986819278766080x^{16}y^{19} - 448782431882821632x^{16}y^{18} \\
& +2077242789707956224x^{16}y^{17} - 8065565248741171200x^{16}y^{16} \\
& +27050338717604757504x^{16}y^{15} - 75661583596283695104x^{16}y^{14} \\
& +174824995253550944256x^{16}y^{13} - 339602162150984951808x^{16}y^{12} \\
& +559609566161055879168x^{16}y^{11} - 767896935069158267904x^{16}y^{10} \\
& +878618703194100440064x^{16}y^9 - 859902478987798771200x^{16}y^8 \\
& +692803462365856963584x^{16}y^7 - 426411802358036474112x^{16}y^6 \\
& +322187026861974898944x^{16}y^5 - 158088614107595213952x^{16}y^4 \\
& -112672111043770677504x^{16}y^3 + 376119230820554151072x^{16}y^2 \\
& -248296823674259391360x^{16}y + 38319779143923607728x^{16} + 129334741106688x^{15}y^{21} \\
& -1290031135653888x^{15}y^{20} - 4534177558413312x^{15}y^{19} + 72807997623386112x^{15}y^{18} \\
& -368649196406562816x^{15}y^{17} + 889454912243171328x^{15}y^{16} \\
& -879063985038200832x^{15}y^{15} + 679714793809145856x^{15}y^{14} \\
& -5420267591081969664x^{15}y^{13} + 24631658843925970944x^{15}y^{12} \\
& -70344359263017409536x^{15}y^{11} + 161399447999941321728x^{15}y^{10} \\
& -319894367917205638656x^{15}y^9 + 559327861622556647424x^{15}y^8 \\
& -852743328954341495040x^{15}y^7 + 1115012049878047811328x^{15}y^6 \\
& -1105437949911372456576x^{15}y^5 + 854505669839873513472x^{15}y^4 \\
& -553095076714553532096x^{15}y^3 + 491639561413665119136x^{15}y^2 \\
& -367913985228651201840x^{15}y - 22528995476102642880x^{15} + 92855711563776x^{14}y^{22} \\
& -2324709064507392x^{14}y^{21} + 29916120500600832x^{14}y^{20} - 214000081469227008x^{14}y^{19} \\
& +1074057041245077504x^{14}y^{18} - 4579013187462537216x^{14}y^{17} \\
& +16975708327233626112x^{14}y^{16} - 52275851083156525056x^{14}y^{15} \\
& +131837608362535428096x^{14}y^{14} - 275660925956543453184x^{14}y^{13} \\
& +494597794084422653952x^{14}y^{12} - 785937619231952919552x^{14}y^{11} \\
& +1103244832323353253888x^{14}y^{10} - 1380189572616846113280x^{14}y^9 \\
& +1598923993737726097920x^{14}y^8 - 1713724561604256198912x^{14}y^7 \\
& +1695298197899541588480x^{14}y^6 - 1505884275861834842496x^{14}y^5 \\
& +1086171214634905779072x^{14}y^4 - 539829461297283894720x^{14}y^3 \\
& +176396061445742500272x^{14}y^2 - 310808348141896630080x^{14}y - 206511595450997105592x^{14} \\
& +195660243966528x^{13}y^{22} - 4612930999471872x^{13}y^{21} + 47934273888239616x^{13}y^{20} \\
& -359571306997923840x^{13}y^{19} + 1958323226070196224x^{13}y^{18} \\
& -8183488271617327104x^{13}y^{17} + 28078379328486371328x^{13}y^{16} \\
& -81048231305396785152x^{13}y^{15} + 194566102846999050240x^{13}y^{14} \\
& -386032750647446863872x^{13}y^{13} + 652232641781862540288x^{13}y^{12} \\
& -975528891419662752768x^{13}y^{11} + 1279615678659214447104x^{13}y^{10} \\
& -1460633970717471744000x^{13}y^9 + 1491707556381535246080x^{13}y^8 \\
& -1342169792975863418112x^{13}y^7 + 1060591299874423219584x^{13}y^6 \\
& -799729857568367764992x^{13}y^5 + 541049959615021847424x^{13}y^4 \\
& -259701243446129095584x^{13}y^3 - 277546570714491120144x^{13}y^2 \\
& -294541286961886794504x^{13}y - 292583529716918323824x^{13} + 19897652477952x^{12}y^{23} \\
& -331627541299200x^{12}y^{22} + 573715646447616x^{12}y^{21} + 17998255735160832x^{12}y^{20} \\
& -242027583122128896x^{12}y^{19} + 1639898606258970624x^{12}y^{18} \\
& -6739904464584523776x^{12}y^{17} + 19969135416280682496x^{12}y^{16} \\
& -49304820667378323456x^{12}y^{15} + 108850846477351053312x^{12}y^{14} \\
& -210889164131245191168x^{12}y^{13} + 349961411826859318272x^{12}y^{12} \\
& -500258677820720901120x^{12}y^{11} + 599222070995018909184x^{12}y^{10} \\
& -589996377600704234496x^{12}y^9 + 477002036807671396608x^{12}y^8 \\
& -249040844151654152448x^{12}y^7 - 24336294362261631360x^{12}y^6 \\
& +180019434685328172288x^{12}y^5 - 181746727495223900832x^{12}y^4 \\
& -68711530724798646816x^{12}y^3 - 614776084975980992736x^{12}y^2 \\
& -318598220416844983752x^{12}y - 109258673535427143744x^{12} + 36479029542912x^{11}y^{23} \\
& -1401126361989120x^{11}y^{22} + 11350781669818368x^{11}y^{21} - 61051801284329472x^{11}y^{20} \\
& +285282177868210176x^{11}y^{19} - 817029115361132544x^{11}y^{18} \\
& +1696190515989590016x^{11}y^{17} - 5149987310479675392x^{11}y^{16} \\
& +19178070561097537536x^{11}y^{15} - 52337731746026889216x^{11}y^{14} \\
& +101597749688026100736x^{11}y^{13} - 160588192623707814912x^{11}y^{12} \\
& +235692417195224418816x^{11}y^{11} - 335468249707895175168x^{11}y^{10}
\end{aligned}$$

$$\begin{aligned}
& +442373905940933714688x^{11}y^9 - 532539208993050411264x^{11}y^8 \\
& +614988554860067543424x^{11}y^7 - 654435369220014446400x^{11}y^6 \\
& +629135347604422294944x^{11}y^5 - 556678570427057004192x^{11}y^4 \\
& -152857924230951167040x^{11}y^3 - 707050900958338605816x^{11}y^2 \\
& -185461955902248705432x^{11}y + 298916031922263507492x^{11} - 13265101651968x^{10}y^{23} \\
& -1576888958877696x^{10}y^{22} + 20046884871536640x^{10}y^{21} - 140905226022617088x^{10}y^{20} \\
& +821934715765800960x^{10}y^{19} - 3632544453784780800x^{10}y^{18} \\
& +11841039514688200704x^{10}y^{17} - 31041842625573765120x^{10}y^{16} \\
& +73249224129007644672x^{10}y^{15} - 159290745829879246848x^{10}y^{14} \\
& +300409293652009998336x^{10}y^{13} - 480738859533916581888x^{10}y^{12} \\
& +670608386660077691904x^{10}y^{11} - 824286472528027124736x^{10}y^{10} \\
& +892823608410242517504x^{10}y^9 - 870798211216547410944x^{10}y^8 \\
& +775426378721644982016x^{10}y^7 - 604090987124793006528x^{10}y^6 \\
& +456302358002275545600x^{10}y^5 - 550672924635607439952x^{10}y^4 \\
& -375932225055060475200x^{10}y^3 - 491401286273932420264x^{10}y^2 \\
& +188018826215615962524x^{10}y + 681159494990613288852x^{10} - 54718544314368x^9y^{23} \\
& -520655239839744x^9y^{22} + 21375053174439936x^9y^{21} - 160986931785990144x^9y^{20} \\
& +807771732545765376x^9y^{19} - 3724549540705124352x^9y^{18} \\
& +13793658747048050688x^9y^{17} - 37787205264953647104x^9y^{16} \\
& +82094775904014041088x^9y^{15} - 157346134988107345920x^9y^{14} \\
& +273508213547292364800x^9y^{13} - 420569356121321011200x^9y^{12} \\
& +567105558413141901312x^9y^{11} - 658432717545772680192x^9y^{10} \\
& +657437408469583690752x^9y^9 - 581932200550032622080x^9y^8 \\
& +446571359110452116160x^9y^7 - 247132961913619488960x^9y^6 \\
& +72249521651790159312x^9y^5 - 419379020670541789584x^9y^4 \\
& -447930546147720634032x^9y^3 - 22241689880634619236x^9y^2 + 606593829516840713088x^9y \\
& +759887379626097800826x^9 + 49744131194880x^8y^{23} - 1820635201732608x^8y^{22} \\
& +15072471752048640x^8y^{21} - 126854167097769984x^8y^{20} + 479803701164802048x^8y^{19} \\
& -1480097657467551744x^8y^{18} + 5710391634686754816x^8y^{17} \\
& -18534789716212408320x^8y^{16} + 41678374845768155136x^8y^{15} \\
& -70231311942250401792x^8y^{14} + 101160678941848276992x^8y^{13} \\
& -135901176904267278336x^8y^{12} + 171773887564444078080x^8y^{11} \\
& -185774117573318630400x^8y^{10} + 166125663295000501248x^8y^9 \\
& -128721078876127163904x^8y^8 + 71190761045140856448x^8y^7 \\
& +22806141852961973376x^8y^6 - 179445321277850624112x^8y^5 \\
& -308369634319809497736x^8y^4 - 247407335958330074160x^8y^3 \\
& +460419242272243115220x^8y^2 + 770015979549195226344x^8y + 475002502674934230450x^8 \\
& +119385914867712x^7y^{23} - 3793819072462848x^7y^{22} + 744503830216704x^7y^{21} \\
& -10880699630026752x^7y^{20} + 109005972825047040x^7y^{19} + 290785951450497024x^7y^{18} \\
& -2698398585007276032x^7y^{17} + 7468375761317117952x^7y^{16} \\
& -14565785493317271552x^7y^{15} + 28573085749555519488x^7y^{14} \\
& -54863900811063705600x^7y^{13} + 89572803477653458944x^7y^{12} \\
& -120210509029091807232x^7y^{11} + 140490354061336209408x^7y^{10} \\
& -144032846610785292288x^7y^9 + 128906815404495928320x^7y^8 \\
& -109311798587636856672x^7y^7 + 9629423777283725504x^7y^6 \\
& -23494457341762169800x^7y^5 - 190795446969025401792x^7y^4 \\
& +67440369703535724672x^7y^3 + 671495699051256920364x^7y^2 + 579081654556007578542x^7y \\
& +56546461281744567072x^7 - 1538751791628288x^6y^{23} + 2520369313873920x^6y^{22} \\
& -17155092711407616x^6y^{21} + 154410757642027008x^6y^{20} - 518898442939711488x^6y^{19} \\
& +1534537634647228416x^6y^{18} - 5693716987375804416x^6y^{17} \\
& +17344764596447133696x^6y^{16} - 38223662137462444032x^6y^{15} \\
& +66427305271718842368x^6y^{14} - 99616930041340643328x^6y^{13} \\
& +135223420889257537536x^6y^{12} - 166931222864578587648x^6y^{11} \\
& +183957571339170241536x^6y^{10} - 176201669810126016000x^6y^9 \\
& +148462420085633074176x^6y^8 - 111370106466365360544x^6y^7 \\
& +56576282173193962944x^6y^6 - 174426813794434848912x^6y^5 \\
& -50661070359680272284x^6y^4 + 25758747468648786880x^6y^3 \\
& +538514788312768522392x^6y^2 + 227474338863617157078x^6y - 201068424441188315172x^6 \\
& -2921638638845952x^5y^{23} + 13450813075095552x^5y^{22} - 31149774954233856x^5y^{21} \\
& +194780607313231872x^5y^{20} - 894546224070967296x^5y^{19} + 2416737779890053120x^5y^{18} \\
& -5456804344543789056x^5y^{17} + 12432363132708311040x^5y^{16} \\
& -25403503499517874176x^5y^{15} + 42620613566523033600x^5y^{14} \\
& -59835038072245309440x^5y^{13} + 74718926351208336384x^5y^{12} \\
& -86480865650211425280x^5y^{11} + 91028734529392479744x^5y^{10} \\
& -84690741425881623552x^5y^9 + 71155888581735910656x^5y^8 \\
& -51860604251401951824x^5y^7 + 5695029581164703280x^5y^6 - 84125744991803559420x^5y^5 \\
& +51197284244830537368x^5y^4 + 242989841133156952170x^5y^3 \\
& +253709997552136022544x^5y^2 - 20942680932223132608x^5y - 214457276725199081694x^5 \\
& +397953049559040x^4y^{23} + 8831241424797696x^4y^{22} - 31952313604177920x^4y^{21} \\
& +96817831613448192x^4y^{20} - 493313378128478208x^4y^{19} + 1615763582872068096x^4y^{18} \\
& -3275302001195630592x^4y^{17} + 4973593377659351040x^4y^{16} \\
& -6865831110902722560x^4y^{15} + 9642338656147445760x^4y^{14}
\end{aligned}$$

$$\begin{aligned}
& -13103636140186472448x^4y^{13} + 15852258006369002496x^4y^{12} \\
& -16944616120747889664x^4y^{11} + 16182835972408392192x^4y^{10} \\
& -14569488788434320384x^4y^9 + 13105346574001727232x^4y^8 - 9946845092554721856x^4y^7 \\
& -13739737281399694080x^4y^6 - 17914824741742793532x^4y^5 \\
& + 71495188566912206532x^4y^4 + 125724995402395510170x^4y^3 \\
& + 47143258601214659508x^4y^2 - 83160518235854048694x^4y - 112737568860336711024x^4 \\
& + 3213470875189248x^3y^{23} - 7327310525005824x^3y^{22} - 5711455330025472x^3y^{21} \\
& + 14948940492914688x^3y^{20} - 9088667303731200x^3y^{19} + 125428997739036672x^3y^{18} \\
& - 374391534551371776x^3y^{17} + 212200794790465536x^3y^{16} + 973524255733831680x^3y^{15} \\
& - 2767241353572790272x^3y^{14} + 4131047543850574848x^3y^{13} \\
& - 4930621858109850624x^3y^{12} + 5650052806707816960x^3y^{11} \\
& - 6131679604372509696x^3y^{10} + 5540427610128799488x^3y^9 - 4095377491195420416x^3y^8 \\
& + 2436695338393533384x^3y^7 - 9903499197480038412x^3y^6 + 7610271609189279384x^3y^5 \\
& + 40672677427679901222x^3y^4 + 33015922564734722250x^3y^3 \\
& - 19241381795928248790x^3y^2 - 49276079762548810338x^3y - 32070076822716449238x^3 \\
& + 1260184656936960x^2y^{23} - 8438262788358144x^2y^{22} + 15841847647862784x^2y^{21} \\
& - 20804653803405312x^2y^{20} + 87465934948810752x^2y^{19} - 321191222574514176x^2y^{18} \\
& + 70572952353574912x^2y^{17} - 1132091446438010880x^2y^{16} + 1672058394343673856x^2y^{15} \\
& - 2488380935774625792x^2y^{14} + 3471185373717288960x^2y^{13} \\
& - 4298841908173780992x^2y^{12} + 4692329333805210624x^2y^{11} \\
& - 4532008196125181952x^2y^{10} + 3920134733432031744x^2y^9 - 3185543561325580800x^2y^8 \\
& + 2037683003864600376x^2y^7 - 2763439487342698644x^2y^6 + 7195643283959171172x^2y^5 \\
& + 11237050534032010962x^2y^4 + 370071084233946024x^2y^3 - 15028017780690408762x^2y^2 \\
& - 14462005786214977134x^2y - 3046257657394496712x^2 - 392978636439552xy^{23} \\
& - 603562125164544xy^{22} + 7874495968149504xy^{21} - 22966865372676096xy^{20} \\
& + 50422309516836864xy^{19} - 130158835546816512xy^{18} + 312883123596115968xy^{17} \\
& + 549628709054201856xy^{16} + 714235769987899392xy^{15} - 805010104195596288xy^{14} \\
& + 951365007057825792xy^{13} - 1149123629691924480xy^{12} + 1222571979370838016xy^{11} \\
& - 1111643499508715520xy^{10} + 948526016220288000xy^9 - 812583803620426752xy^8 \\
& + 522714873253813632xy^7 - 139123043184240432xy^6 + 1995240314351114496xy^5 \\
& + 817153520782879764xy^4 - 2054764273424057316xy^3 - 3805628737924244844xy^2 \\
& - 1857635884100264796xy + 892049845208691312x - 175762596888576y^{23} + 722948040032256y^{22} \\
& - 295148511756288y^{21} - 5024157250682880y^{20} + 16359186612289536y^{19} \\
& - 26520254477697024y^{18} + 32136366889598976y^{17} - 49377682761744384y^{16} \\
& + 92478485235548160y^{15} - 138739698177933312y^{14} + 152964447027535872y^{13} \\
& - 135697430020939776y^{12} + 110372900096839680y^{11} - 90633807037071360y^{10} \\
& + 82322909951293440y^9 - 77639292732082176y^8 + 51912176633611128y^7 + 43120064495391912y^6 \\
& + 144217372785139368y^5 - 146656034104891572y^4 - 335435973077817468y^3 \\
& - 299537345387594916y^2 + 2874079988932608y + 222260390599949388) \\
i = 11, D_i = 1, \varphi_i = 1, e_i = 1, \rho_{ij}: 9:1, 10:1, 26:1, 28:2, 29:1, 35:3 \\
w_i \quad \langle -4t^2 + 8t - 20, \\
\quad -4t^2 + 8t - 20 \rangle \\
i = 12, D_i = 128, \varphi_i = 1, \rho_{ij}: 4:3, 7:3, 9:2, 10:3, 14:2, 16:1, 21:1, 25:1, 28:2, 31:1, 39:1, 40:3, 43:3, 46:1 \\
w_i \quad \langle 18029t^3 + 21934t^2 + 7767t - 662, \\
\quad -13929t^3 + 4486t^2 + 18091t - 2914 \rangle \\
i = 13, D_i = 1, \varphi_i = 1, \rho_{ij}: 0:1, 5:2, 10:1, 16:2, 24:2, 31:3, 38:3, 40:3, 44:3 \\
w_i \quad \langle 4t^3 + 8t^2 + 22t + 18, \\
\quad -6t^3 - 20t^2 - 12t - 10 \rangle \\
i = 14, D_i = 37, \varphi_i = \tau(\Pi_{15}), \rho_{ij}: 1:3, 4:3, 5:2, 12:2, 22:1, 23:1, 33:1, 34:2, 36:3, 44:2 \\
w_i \quad \langle -23t^3 - 26t^2 + 62t + 107, \\
\quad 10t^3 - 8t^2 - 72t + 50 \rangle \\
i = 16, D_i = 1, \varphi_i = 1, \rho_{ij}: 4:2, 5:3, 6:1, 8:2, 12:3, 13:2, 22:3, 24:2, 27:1, 38:3, 43:3, 44:1, 46:2 \\
w_i \quad \langle -26t^3 + 10t^2 + 59t - 21, \\
\quad -28t^3 + 90t^2 - 73t + 57 \rangle \\
i = 17, D_i = 1, \varphi_i = 1, \rho_{ij}: 5:1, 7:2, 15:1, 23:1, 32:3, 40:2, 43:3, 46:1 \\
w_i \quad \langle 11t^3 - 18t^2 + 11t, \\
\quad -3t^3 - 6t^2 - t - 2 \rangle \\
i = 18, D_i = 1, \varphi_i = 1, \rho_{ij}: 23:3, 25:1, 31:3, 38:3, 40:1, 45:1 \\
w_i \quad \langle 2t^3 - 6t^2 + 4t, \\
\quad -t^3 - 2t^2 - 8t + 1 \rangle \\
i = 19, D_i = 1, \varphi_i = 1, \rho_{ij}: 3:2, 9:1, 10:1, 20:1, 25:1, 26:3, 27:2, 30:1, 31:2, 38:3, 39:2, 41:1, 45:3 \\
w_i \quad \langle 13t^3 + 5t^2 - 7t - 11, \\
\quad -13t^3 + 3t^2 - 29t + 15 \rangle \\
i = 20, D_i = 4, \varphi_i = 1, e_i = 4, \rho_{ij}: 2:1, 7:2, 15:2, 19:3, 33:1, 38:3, 45:3 \\
w_i \quad \langle -8t^3 - 24t^2 - 24t - 8, \\
\quad -8t^3 - 8t^2 - 24t - 24 \rangle \\
i = 21, D_i = 1, \varphi_i = \tau(\Pi_{22}), \rho_{ij}: 0:3, 1:1, 3:1, 4:3, 7:2, 9:2, 12:3, 25:3, 27:1, 32:2, 34:3, 38:3, 39:1, 43:1, 44:2, 45:2 \\
w_i \quad \langle -172t^3 - 188t^2 - 206t - 118, \\
\quad -66t^3 - 120t^2 - 56t - 146 \rangle \\
i = 23, D_i = 1, \varphi_i = 1, \rho_{ij}: 9:1, 14:3, 15:3, 17:3, 18:1, 24:2, 31:1, 36:3, 44:1, 45:3, 46:3
\end{aligned}$$

$$\begin{aligned}
w_i & \langle 31t^5 + 105t^4 - 35t^3 - 231t^2 - 83t + 17, \\
& \quad -17t^5 + 45t^4 + 215t^3 + 57t^2 - 139t + 11 \rangle \\
i = 24, D_i = 1, \varphi_i = 1, \rho_{ij}: & 13:2, 16:2, 22:2, 23:2, 35:2, 39:2 \\
w_i & \langle -2t^3 + 4t^2, \\
& \quad -t^3 + 2t^2 - 5t \rangle \\
i = 25, D_i = 1, \varphi_i = 1, \rho_{ij}: & 8:1, 9:3, 12:3, 15:3, 18:3, 19:3, 21:1, 22:1, 26:1, 32:1, 37:3, 38:1, 40:1, 41:1, 45:1 \\
w_i & \langle 256t^5 + 850t^4 + 1310t^3 + 682t^2 - 340t - 2, \\
& \quad -294t^5 - 490t^4 + 558t^3 + 1390t^2 + 1052t + 652 \rangle \\
i = 26, D_i = 1, \varphi_i = 1, \rho_{ij}: & 5:1, 8:2, 10:3, 11:3, 15:3, 19:1, 22:3, 25:3, 31:1, 34:1, 37:1, 40:2, 41:1, 45:3 \\
w_i & \langle 22t^5 + 70t^4 + 72t^3 + 80t^2 + 110t + 54, \\
& \quad 10t^5 - 2t^4 + 44t^3 + 108t^2 + 118t + 66 \rangle \\
i = 27, D_i = 1, \varphi_i = \tau(\Pi_{35}), \rho_{ij}: & 16:3, 19:2, 21:3, 28:2, 29:2, 30:3, 31:1, 38:1, 40:3, 46:2 \\
w_i & \langle 5t^3 - 18t^2 + 11t + 2, \\
& \quad 3t^3 - 8t^2 + 11t + 2 \rangle \\
i = 28, D_i = 1, \varphi_i = 1, \rho_{ij}: & 5:2, 9:2, 11:2, 12:2, 27:2, 30:2, 33:2 \\
w_i & \langle -10t^3 - 18t^2 + 26t + 2, \\
& \quad 12t^3 - 34t^2 - 16t - 2 \rangle \\
i = 29, D_i = 1, \varphi_i = 1, \rho_{ij}: & 5:2, 11:3, 15:2, 27:2, 30:2, 33:1, 34:1, 38:3, 39:3, 46:3 \\
w_i & \langle 8t^3 - 25t^2 - 45t - 6, \\
& \quad 14t^3 + 5t^2 + 35t + 22 \rangle \\
i = 30, D_i = 77367, \varphi_i = \tau(\Pi_{38}), \rho_{ij}: & 3:2, 8:1, 9:3, 19:3, 27:1, 28:2, 29:2, 44:1 \\
w_i & \langle -20168t^2 - 98284t - 316358, \\
& \quad -35132t^2 - 59387t + 87553 \rangle \\
i = 31, D_i = 74, \varphi_i = \tau(\Pi_{41}), \rho_{ij}: & 1:1, 5:1, 12:3, 13:1, 18:1, 19:2, 23:3, 26:3, 27:3, 37:3, 45:2, 46:1 \\
w_i & \langle 88t^3 + 293t^2 + 354t + 666, \\
& \quad -471t^3 - 536t^2 + 52t - 777 \rangle \\
i = 32, D_i = 2, \varphi_i = \tau(\Pi_{42}), \rho_{ij}: & 5:3, 8:3, 17:1, 21:2, 25:3 \\
W & \langle 2y^2 - 2y + 14, \\
& \quad 2y^2 + 5y - 7 \rangle \\
A & \langle -384y^9 + 2112y^8 - 1248y^7 - 272y^6 - 8520y^5 + 3756y^4 + 14602y^3 - 444y^2 + 39216y - 31640, \\
& \quad 384y^9 - 192y^8 - 672y^7 - 5840y^6 + 11208y^5 - 5796y^4 + 31190y^3 - 34704y^2 + 8496y - 30224 \rangle \\
B & \langle -384x^{17} + 1920x^{16}y + 1152x^{16} - 2880x^{15}y^2 + 1728x^{15}y + 7872x^{15} + 1152x^{14}y^3 \\
& \quad - 3648x^{14}y^2 - 13056x^{14}y + 768x^{14} + 1920x^{13}y^4 + 864x^{13}y^3 + 23808x^{13}y^2 \\
& \quad + 2304x^{13}y - 24000x^{13} - 1536x^{12}y^5 + 192x^{12}y^4 - 35808x^{12}y^3 + 1440x^{12}y^2 \\
& \quad + 52800x^{12}y - 11328x^{12} + 192x^{11}y^6 + 10464x^{11}y^5 + 29088x^{11}y^4 - 6096x^{11}y^3 \\
& \quad - 80544x^{11}y^2 + 9888x^{11}y + 28416x^{11} - 8640x^{10}y^6 - 5664x^{10}y^5 - 22560x^{10}y^4 \\
& \quad + 79296x^{10}y^3 - 14544x^{10}y^2 - 79584x^{10}y + 17952x^{10} + 1152x^9y^7 + 21216x^9y^6 \\
& \quad + 42480x^9y^5 - 10008x^9y^4 - 15000x^9y^3 + 129024x^9y^2 - 39984x^9y - 14112x^9 \\
& \quad - 20160x^8y^7 - 2016x^8y^6 + 66240x^8y^5 - 5472x^8y^4 - 158712x^8y^3 \\
& \quad + 80520x^8y^2 + 43200x^8y - 10800x^8 + 2880x^7y^8 + 17952x^7y^7 + 8976x^7y^6 \\
& \quad - 40536x^7y^5 + 91512x^7y^4 - 72108x^7y^3 - 56568x^7y^2 + 33432x^7y + 1824x^7 \\
& \quad - 24960x^6y^8 + 12192x^6y^7 + 78048x^6y^6 - 101304x^6y^5 + 6696x^6y^4 \\
& \quad + 93360x^6y^3 - 66588x^6y^2 - 4488x^6y + 2232x^6 + 3840x^5y^9 + 1632x^5y^8 \\
& \quad - 12816x^5y^7 + 74424x^5y^6 - 31788x^5y^5 - 99828x^5y^4 + 110502x^5y^3 \\
& \quad - 10560x^5y^2 - 8244x^5y + 456x^5 - 17280x^4y^9 + 13152x^4y^8 + 12288x^4y^7 \\
& \quad + 1608x^4y^6 + 30672x^4y^5 - 93672x^4y^4 + 45006x^4y^3 + 5550x^4y^2 - 1584x^4y \\
& \quad + 108x^4 + 2880x^3y^{10} - 7200x^3y^9 - 3600x^3y^8 + 43512x^3y^7 - 29900x^3y^6 \\
& \quad + 32586x^3y^5 - 43140x^3y^4 + 2038x^3y^3 - 1236x^3y^2 - 19086x^3y + 15796x^3 \\
& \quad - 6336x^2y^{10} + 2784x^2y^9 - 7488x^2y^8 + 30696x^2y^7 - 17024x^2y^6 \\
& \quad + 12414x^2y^5 - 8124x^2y^4 - 6948x^2y^3 + 5514x^2y^2 - 11094x^2y + 15388x^2 \\
& \quad + 1152xy^{11} - 4704xy^{10} + 2160xy^9 + 3048xy^8 + 8316xy^7 - 8354xy^6 - 651xy^5 + 2268xy^4 \\
& \quad + 1638xy^3 + 6678xy^2 + 18057xy - 8330x - 960y^{11} - 1248y^{10} + 96y^9 + 4456y^8 - 408y^7 \\
& \quad - 8506y^6 + 5310y^5 + 139y^4 + 20628y^3 + 1504y^2 - 34440y - 14683, \\
& \quad -384x^{17} - 2688x^{16} + 2112x^{15}y^2 + 6720x^{15}y + 960x^{15} - 4416x^{14}y^3 - 7488x^{14}y^2 \\
& \quad + 1920x^{14}y + 15360x^{14} + 3072x^{13}y^4 + 8160x^{13}y^3 + 3264x^{13}y^2 - 28800x^{13}y \\
& \quad + 4800x^{13} - 192x^{12}y^5 - 15744x^{12}y^4 - 2112x^{12}y^3 + 59040x^{12}y^2 - 768x^{12}y \\
& \quad - 29376x^{12} - 192x^{11}y^6 + 13920x^{11}y^5 - 7392x^{11}y^4 - 53424x^{11}y^3 \\
& \quad + 15744x^{11}y^2 + 69408x^{11}y - 16320x^{11} - 768x^{10}y^6 - 10368x^{10}y^5 + 31680x^{10}y^4 \\
& \quad + 22032x^{10}y^3 - 102192x^{10}y^2 + 20736x^{10}y + 22944x^{10} - 1152x^9y^7 + 24672x^9y^6 \\
& \quad - 45552x^9y^5 - 45648x^9y^4 + 153384x^9y^3 - 42096x^9y^2 - 72912x^9y + 16320x^9 \\
& \quad - 1152x^8y^7 + 23808x^8y^6 + 11664x^8y^5 - 126000x^8y^4 + 58272x^8y^3 \\
& \quad + 100104x^8y^2 - 45840x^8y - 5904x^8 - 2880x^7y^8 + 21408x^7y^7 - 33744x^7y^6 \\
& \quad + 37224x^7y^5 + 216x^7y^4 - 108660x^7y^3 + 73152x^7y^2 + 19416x^7y - 5424x^7 \\
& \quad - 960x^6y^8 + 41664x^6y^7 - 40944x^6y^6 - 69408x^6y^5 + 131184x^6y^4 \\
& \quad - 82788x^6y^3 - 30948x^6y^2 + 22560x^6y - 168x^6 - 3840x^5y^9 + 9696x^5y^8 \\
& \quad + 18192x^5y^7 + 20184x^5y^6 - 119988x^5y^5 + 99108x^5y^4 + 24822x^5y^3 \\
& \quad - 49764x^5y^2 + 5460x^5y + 144x^5 - 960x^4y^9 + 22464x^4y^8 - 20976x^4y^7 \\
& \quad + 23568x^4y^6 - 67716x^4y^5 + 17292x^4y^4 + 49704x^4y^3 - 19602x^4y^2 + 1476x^4y \\
& \quad - 156x^4 - 2880x^3y^{10} + 2784x^3y^9 + 26256x^3y^8 - 14376x^3y^7 - 4532x^3y^6 \\
& \quad - 16206x^3y^5 - 9972x^3y^4 + 4034x^3y^3 + 13032x^3y^2 - 3102x^3y + 14980x^3 \rangle
\end{aligned}$$

$ \begin{aligned} & -1152x^2y^{10} + 3840x^2y^9 + 10320x^2y^8 - 1296x^2y^7 - 16692x^2y^6 \\ & + 2868x^2y^5 - 5304x^2y^4 - 3610x^2y^3 + 8286x^2y^2 + 7992x^2y - 408x^2 - 1152xy^{11} \\ & + 1056xy^{10} + 6480xy^9 + 168xy^8 + 2100xy^7 - 10818xy^6 + 11199xy^5 - 2448xy^4 + 16112xy^3 \\ & - 15966xy^2 + 13791xy - 23010x - 768y^{11} + 960y^{10} + 4656y^9 - 4992y^8 + 460y^7 - 5640y^6 \\ & + 11777y^5 + 840y^4 - 13064y^3 - 11034y^2 - 41789y + 35172) \end{aligned} $		
$i = 33$,	$D_i = 1$,	$\varphi_i = 1$, ρ_{ij} : 0:2, 8:2, 9:2, 14:3, 20:3, 28:2, 29:3, 34:2, 35:3, 36:2, 41:2, 44:1
w_i	$ \begin{aligned} & \langle 54t^2 + 342t + 5040, \\ & -90t^2 - 774t - 1260 \rangle \end{aligned} $	
$i = 34$,	$D_i = 2050$,	$\varphi_i = \tau(\Pi_{44})$, ρ_{ij} : 1:2, 4:3, 5:2, 6:1, 9:3, 14:2, 21:1, 26:3, 29:3, 33:2, 36:3, 37:3, 45:1
w_i	$ \begin{aligned} & \langle 4139t^5 - 8564t^4 - 2955t^3 + 24185t^2 - 14918t - 15699, \\ & -1248t^5 - 7827t^4 + 25885t^3 - 4145t^2 - 16674t + 2168 \rangle \end{aligned} $	
$i = 36$,	$D_i = 1$,	$\varphi_i = 1$, ρ_{ij} : 2:1, 4:3, 6:3, 14:1, 23:1, 33:2, 34:1, 37:2, 46:2
w_i	$ \begin{aligned} & \langle t^3 + 3t^2 - 4t - 2, \\ & 4t^3 - t^2 - 5t \rangle \end{aligned} $	
$i = 37$,	$D_i = 1$,	$\varphi_i = 1$, ρ_{ij} : 0:1, 3:2, 4:3, 7:1, 8:2, 10:3, 15:2, 25:1, 26:3, 31:1, 34:1, 36:2, 40:2, 44:2
w_i	$ \begin{aligned} & \langle 36t^2 - 78t + 450, \\ & 42t^2 - 198t + 384 \rangle \end{aligned} $	
$i = 39$,	$D_i = 1$,	$\varphi_i = 1$, ρ_{ij} : 0:2, 3:2, 9:2, 10:3, 12:3, 19:2, 21:3, 22:1, 24:2, 29:1, 40:1, 41:1
w_i	$ \begin{aligned} & \langle -2t^2 - 10t + 16, \\ & -14t^2 + 26t - 12 \rangle \end{aligned} $	
$i = 40$,	$D_i = 1$,	$\varphi_i = 1$, ρ_{ij} : 2:1, 4:3, 6:3, 9:2, 12:1, 13:1, 17:2, 18:3, 25:3, 26:2, 27:1, 37:2, 38:1, 39:3
w_i	$ \begin{aligned} & \langle -134t^3 + 90t^2 + 106t - 102, \\ & -76t^3 - 172t^2 + 124t + 132 \rangle \end{aligned} $	
$i = 43$,	$D_i = 1$,	$\varphi_i = 1$, ρ_{ij} : 5:3, 12:1, 15:1, 16:1, 17:1, 21:3, 41:3
w_i	$ \begin{aligned} & \langle 2t^2 + 16t + 44, \\ & -4t^2 - 14t + 56 \rangle \end{aligned} $	
$i = 45$,	$D_i = 576554341310116$,	φ_i below, ρ_{ij} : 9:2, 18:3, 19:1, 20:1, 21:2, 23:1, 25:3, 26:1, 31:2, 34:3
φ_i	$ \begin{aligned} & -3056125324896045042356373657905152y^2 + 5808087145261556140762007367192576y \\ & - 3735494414013855498981142293907456 \end{aligned} $	
w_i	$ \begin{aligned} & \langle -4617605t^2 + 6464647t + 22164504, \\ & 10158731t^2 + 923521t + 25858588 \rangle \end{aligned} $	
$i = 46$,	$D_i = 1041267180625$,	φ_i below, ρ_{ij} : 8:2, 9:3, 12:3, 16:2, 17:3, 23:1, 27:2, 29:1, 31:3, 36:2
φ_i	$ \begin{aligned} & 29318645550557691451617200000000y^2 - 55684784467607883897663200000000y \\ & + 70145841845090532050143600000000 \end{aligned} $	
w_i	$ \begin{aligned} & \langle -33614t^2 - 31213t - 797132, \\ & 175273t^2 - 201684t + 657874 \rangle \end{aligned} $	

Table 3: Edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(0, 8)	0	$2^4 \cdot 3^2 \cdot 5$	(14, 36)	14	2
(0, 9)	0	$2^4 \cdot 5 \cdot 13$	(14, 44)	14	$2^2 \cdot 3^4 \cdot 5$
(0, 13)	0	2^6	(15, 17)	15	$2 \cdot 3^4 \cdot 13$
(0, 21)	0	$5 \cdot 13$	(15, 20)	15	2^5
(0, 33)	0	$2^3 \cdot 3^2 \cdot 13$	(15, 23)	15	$2^4 \cdot 3^4 \cdot 13$
(0, 37)	0	$3^2 \cdot 5$	(15, 25)	15	$2 \cdot 5 \cdot 37^2$
(0, 39)	0	2^3	(15, 26)	15	$2^2 \cdot 5^2$
(1, 8)	1	$2^3 \cdot 13 \cdot 17$	(15, 29)	15	$3^2 \cdot 5$
(1, 9)	1	$2^3 \cdot 5 \cdot 13 \cdot 17$	(15, 37)	15	37
(1, 14)	1	$13 \cdot 17$	(15, 43)	15	$2 \cdot 13 \cdot 37$
(1, 21)	1	$13 \cdot 17$	(16, 22)	16	$2^2 \cdot 5$
(1, 31)	1	$5 \cdot 13$	(16, 24)	16	$3^2 \cdot 5$
(1, 34)	1	$5^3 \cdot 17^2$	(16, 27)	16	$5^2 \cdot 17$
(2, 9)	2	$2^4 \cdot 13^2$	(16, 38)	16	$2^2 \cdot 3^2 \cdot 17$
(2, 20)	2	2^6	(16, 43)	16	$2^6 \cdot 17$
(2, 36)	2	2^2	(16, 44)	16	$2^2 \cdot 5^2 \cdot 17$
(2, 40)	2	$2^4 \cdot 3^2$	(16, 46)	16	$2^2 \cdot 5$
(3, 8)	3	2^6	(17, 23)	17	$2^9 \cdot 3^4$
(3, 19)	3	$3^2 \cdot 17$	(17, 32)	17	$2^6 \cdot 13 \cdot 61$
(3, 21)	3	$2^4 \cdot 3^2$	(17, 40)	17	2
(3, 30)	3	$2^3 \cdot 3^2 \cdot 17$	(17, 43)	17	$2^{11} \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(3, 37)	3	3^4	(17, 46)	17	2^8
(3, 39)	3	2	(18, 23)	18	2
(4, 12)	4	$2^3 \cdot 3^2 \cdot 5^2$	(18, 25)	18	$3^2 \cdot 5 \cdot 41$
(4, 14)	4	$2 \cdot 3^2 \cdot 5$	(18, 31)	18	$3^4 \cdot 5 \cdot 13$
(4, 16)	4	5^2	(18, 38)	18	$2^2 \cdot 17 \cdot 41$
(4, 21)	4	$3^4 \cdot 5$	(18, 40)	18	$3^2 \cdot 17$
(4, 34)	4	$3^8 \cdot 5$	(18, 45)	18	$2^3 \cdot 13$
(4, 36)	4	$2 \cdot 3^2$	(19, 20)	19	2^8
(4, 37)	4	$3^2 \cdot 5$	(19, 25)	19	$2^6 \cdot 7^2$
(4, 40)	4	$2 \cdot 3^2$	(19, 26)	19	$2^2 \cdot 7^2 \cdot 37$
(5, 13)	5	$2^6 \cdot 13$	(19, 27)	19	2^4
(5, 14)	5	$2^3 \cdot 7^2 \cdot 13$	(19, 30)	19	$2^7 \cdot 17$
(5, 16)	5	$5^2 \cdot 17$	(19, 31)	19	2^4
(5, 17)	5	$2^3 \cdot 13 \cdot 61$	(19, 38)	19	$2^5 \cdot 3^2$
(5, 26)	5	$2^6 \cdot 3^2$	(19, 39)	19	2^3
(5, 28)	5	$2^4 \cdot 13^2 \cdot 17$	(19, 41)	19	$2^3 \cdot 3^4$
(5, 29)	5	$5^4 \cdot 13$	(19, 45)	19	$2^3 \cdot 13$
(5, 31)	5	$2^3 \cdot 5^2 \cdot 13$	(20, 33)	20	$2^5 \cdot 3^2$
(5, 32)	5	$2^6 \cdot 13 \cdot 61$	(20, 38)	20	$2^4 \cdot 3^2 \cdot 13^2$
(5, 34)	5	$7^2 \cdot 17^2$	(20, 45)	20	2^4
(5, 43)	5	$2^4 \cdot 13 \cdot 17$	(21, 25)	21	$13 \cdot 37$
(6, 7)	6	$2 \cdot 3^2 \cdot 13$	(21, 27)	21	$13 \cdot 17 \cdot 37$
(6, 16)	6	$2 \cdot 3^2 \cdot 17$	(21, 32)	21	$2^4 \cdot 13^2$
(6, 34)	6	$2^8 \cdot 17$	(21, 34)	21	$2^4 \cdot 3^6$
(6, 36)	6	1	(21, 38)	21	$2^2 \cdot 3^2 \cdot 37$
(6, 40)	6	$2^2 \cdot 13$	(21, 39)	21	2
(7, 12)	7	2	(21, 43)	21	$2^7 \cdot 13$
(7, 17)	7	$2^6 \cdot 13^2$	(21, 44)	21	$2^2 \cdot 3^2 \cdot 5$
(7, 20)	7	$2^4 \cdot 13$	(21, 45)	21	2^3
(7, 21)	7	2^7	(22, 24)	22	17
(7, 37)	7	$3^2 \cdot 13$	(22, 25)	22	5^2
(7, 38)	7	$2^2 \cdot 13$	(22, 26)	22	$3^2 \cdot 5^3$
(8, 9)	8	$2^{11} \cdot 5 \cdot 13 \cdot 17$	(22, 39)	22	2
(8, 16)	8	$2^{11} \cdot 3^2 \cdot 5$	(23, 24)	23	5^6
(8, 25)	8	$2^{10} \cdot 3^4 \cdot 5^2$	(23, 31)	23	$5^4 \cdot 13$
(8, 26)	8	$2^{10} \cdot 5^3$	(23, 36)	23	3^2
(8, 30)	8	$2^6 \cdot 3^2 \cdot 13$	(23, 44)	23	$2^6 \cdot 3^2 \cdot 5^3$
(8, 32)	8	$2^7 \cdot 3^2$	(23, 45)	23	2^2
(8, 33)	8	$2^4 \cdot 3^2 \cdot 13 \cdot 17$	(23, 46)	23	$2^4 \cdot 17$
(8, 37)	8	$2^3 \cdot 3^2 \cdot 13$	(24, 35)	24	$2 \cdot 3^2 \cdot 17$
(8, 42)	8	$2^7 \cdot 3^2 \cdot 13^2$	(24, 39)	24	$2 \cdot 5^2$
(8, 44)	8	$2^{11} \cdot 3^4 \cdot 17$	(25, 26)	25	$2 \cdot 3^2 \cdot 5^3 \cdot 7^2$
(8, 46)	8	$2^7 \cdot 5$	(25, 32)	25	$2^6 \cdot 3^2 \cdot 13^2$
(9, 11)	11	$2^{10} \cdot 5 \cdot 13$	(25, 37)	25	$3^2 \cdot 13$
(9, 12)	12	$2^5 \cdot 5 \cdot 13 \cdot 17$	(25, 38)	25	$2^4 \cdot 3^4 \cdot 41$
(9, 19)	19	$2^4 \cdot 3^2 \cdot 17$	(25, 40)	25	$2 \cdot 3^2 \cdot 13$
(9, 21)	21	$2^4 \cdot 13^2$	(25, 41)	25	$2 \cdot 3^4 \cdot 37$
(9, 23)	23	$2^6 \cdot 5 \cdot 13 \cdot 17$	(25, 45)	25	2^2
(9, 25)	25	$2^6 \cdot 3^4 \cdot 5 \cdot 13$	(26, 31)	26	$2 \cdot 3^4 \cdot 5^2$
(9, 28)	28	$2^4 \cdot 5 \cdot 13 \cdot 17$	(26, 34)	26	5^7
(9, 30)	30	$2^3 \cdot 5^2 \cdot 13 \cdot 17$	(26, 37)	26	3^4
(9, 33)	33	$2^3 \cdot 3^2 \cdot 17 \cdot 29$	(26, 40)	26	$2^2 \cdot 3^2$
(9, 34)	34	$2^6 \cdot 5^2 \cdot 13 \cdot 29$	(26, 41)	26	$2 \cdot 5^3 \cdot 13$
(9, 39)	39	$2^3 \cdot 5$	(26, 45)	26	$2 \cdot 13$
(9, 40)	40	$2^4 \cdot 5 \cdot 13$	(27, 28)	27	2^8
(9, 44)	44	$2^6 \cdot 5^2 \cdot 17$	(27, 29)	27	$5^3 \cdot 13$
(9, 45)	45	$2^3 \cdot 13$	(27, 30)	27	$2^3 \cdot 3^2 \cdot 5^2$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(9, 46)	46	$2^3 \cdot 5 \cdot 17$	(27, 31)	27	$2^4 \cdot 5^3$
(10, 11)	10	$2^{10} \cdot 5 \cdot 13$	(27, 38)	27	$2^3 \cdot 17 \cdot 37$
(10, 12)	10	$2^{11} \cdot 3^2 \cdot 5^2$	(27, 40)	27	$2 \cdot 5^2$
(10, 13)	10	$2^{10} \cdot 13 \cdot 29$	(27, 46)	27	2^4
(10, 19)	10	$2^{10} \cdot 13 \cdot 37$	(28, 30)	28	$2^2 \cdot 13^2$
(10, 26)	10	$2^{14} \cdot 3^2 \cdot 5^2 \cdot 37$	(28, 33)	28	$2 \cdot 17$
(10, 37)	10	$2^3 \cdot 3^2 \cdot 5^2$	(29, 30)	29	$2^3 \cdot 5 \cdot 13$
(10, 39)	10	$2^3 \cdot 13$	(29, 33)	29	$2 \cdot 3^2 \cdot 13$
(11, 26)	11	$2^{14} \cdot 5^4$	(29, 34)	29	$2 \cdot 3^4 \cdot 5^3$
(11, 28)	11	$2^{10} \cdot 5^3 \cdot 13$	(29, 38)	29	$2^3 \cdot 3^2 \cdot 13$
(11, 29)	11	$2^4 \cdot 5^3 \cdot 13$	(29, 39)	29	$2 \cdot 5 \cdot 13$
(11, 35)	11	$2^{10} \cdot 5^3 \cdot 13$	(29, 46)	29	$2^4 \cdot 5$
(12, 14)	12	$2 \cdot 5^2$	(30, 44)	30	$2^2 \cdot 3^4 \cdot 5^2$
(12, 16)	12	$2 \cdot 5$	(31, 37)	31	3^4
(12, 21)	12	$2 \cdot 3^2 \cdot 5^2 \cdot 13$	(31, 45)	31	$2 \cdot 13$
(12, 25)	12	$2^4 \cdot 3^2 \cdot 13^2$	(31, 46)	31	$2 \cdot 5^2$
(12, 28)	12	$2^5 \cdot 5^2 \cdot 17$	(33, 34)	33	$2 \cdot 3^4 \cdot 29$
(12, 31)	12	$2 \cdot 3^4 \cdot 5$	(33, 35)	33	$2 \cdot 3^2 \cdot 17$
(12, 39)	12	$2^5 \cdot 5$	(33, 36)	33	$2^2 \cdot 3^2 \cdot 13$
(12, 40)	12	$2 \cdot 3^4 \cdot 5$	(33, 41)	33	$2^3 \cdot 3^4$
(12, 43)	12	$2^{11} \cdot 13$	(33, 44)	33	$2 \cdot 3^2 \cdot 17$
(12, 46)	12	$2^5 \cdot 5^2$	(34, 36)	34	3^4
(13, 16)	13	5^3	(34, 37)	34	3^2
(13, 24)	13	$2^2 \cdot 5$	(34, 45)	34	$2^3 \cdot 13$
(13, 31)	13	$2^2 \cdot 5$	(36, 37)	36	13
(13, 38)	13	$2^2 \cdot 13^2$	(36, 46)	36	2
(13, 40)	13	$2^4 \cdot 5 \cdot 13$	(37, 40)	37	$3^2 \cdot 13$
(13, 44)	13	$2^2 \cdot 5^2 \cdot 29$	(37, 44)	37	$3^4 \cdot 5 \cdot 37$
(14, 22)	14	$3^2 \cdot 37$	(38, 40)	38	$2 \cdot 13 \cdot 17$
(14, 23)	14	$2^4 \cdot 13$	(39, 40)	39	$2 \cdot 5^2$
(14, 33)	14	$2 \cdot 13 \cdot 17$	(39, 41)	39	2^4
(14, 34)	14	$2^2 \cdot 3^2 \cdot 7^2 \cdot 37$	(41, 43)	41	$2 \cdot 13 \cdot 37$

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	5	2	(1, 4), (9, 28), (17, 20), (25, 12)
2	6	0	(3, 22), (5, 9), (5, 41), (5, 48), (11, 14), (13, 21), (13, 40), (13, 53), (19, 6), (21, 17), (21, 32), (21, 49), (27, 62), (29, 24), (29, 29), (29, 61), (35, 54), (37, 16), (37, 25), (37, 57), (43, 46), (45, 5), (45, 8), (45, 37), (51, 38), (53, 0), (53, 1), (53, 33), (59, 30), (61, 13), (61, 45), (61, 56)
2	6	2	(7, 34), (15, 26), (23, 18), (31, 10), (39, 2), (47, 58), (55, 50), (63, 42)
2	6	3	(5, 7), (5, 39), (13, 3), (13, 35), (21, 15), (21, 47), (29, 11), (29, 43), (37, 23), (37, 55), (45, 19), (45, 51), (53, 31), (53, 63), (61, 27), (61, 59)
3	1	0	(0, 2)
3	2	0	(0, 3), (0, 4), (3, 6), (3, 7), (6, 0), (6, 1)
5	3	0	(1, 117), (3, 112), (6, 22), (13, 22), (16, 57), (18, 2), (21, 62), (23, 82), (26, 17), (31, 47), (38, 47), (41, 82), (43, 27), (46, 87), (48, 107), (51, 42), (56, 72), (63, 72), (66, 107), (68, 52), (71, 112), (73, 7), (76, 67), (81, 97), (86, 77), (88, 97), (91, 7), (93, 77), (98, 32), (101, 92), (106, 122), (113, 122), (116, 32), (118, 102), (121, 37)

p	k	B_p	classes (a, b)
5	4	0	(8, 392), (11, 127), (28, 262), (33, 167), (36, 527), (53, 162), (58, 567), (61, 302), (78, 62), (83, 342), (96, 512), (103, 587), (108, 117), (111, 477), (123, 57), (133, 517), (136, 252), (153, 387), (158, 292), (161, 27), (178, 287), (183, 67), (186, 427), (203, 187), (208, 467), (221, 12), (228, 87), (233, 242), (236, 602), (248, 182), (258, 17), (261, 377), (278, 512), (283, 417), (286, 152), (303, 412), (308, 192), (311, 552), (328, 312), (333, 592), (346, 137), (353, 212), (358, 367), (361, 102), (373, 307), (383, 142), (386, 502), (403, 12), (408, 542), (411, 277), (428, 537), (433, 317), (436, 52), (453, 437), (458, 92), (471, 262), (478, 337), (483, 492), (486, 227), (498, 432), (508, 267), (511, 2), (528, 137), (533, 42), (536, 402), (553, 37), (561, 177), (578, 562), (583, 217), (596, 387), (603, 462), (608, 617), (611, 352), (623, 557)
5	5	0	(558, 442), (1183, 1067), (1808, 1692), (2433, 2317), (3058, 2942)
7	1	0	(0, 4), (1, 3), (2, 0), (2, 1), (2, 6), (3, 3), (5, 4)
13	1	3	(6, 5)
13	2	3	(0, 8), (2, 76), (5, 37), (7, 10), (7, 23), (7, 36), (7, 49), (7, 62), (7, 75), (7, 88), (7, 97), (7, 101), (7, 114), (7, 127), (7, 140), (7, 153), (7, 166), (9, 141), (10, 37), (11, 111), (12, 163), (13, 138), (15, 154), (18, 154), (20, 6), (22, 154), (23, 115), (24, 98), (25, 150), (26, 99), (28, 63), (31, 102), (33, 84), (35, 167), (36, 24), (37, 85), (38, 137), (39, 60), (41, 141), (44, 50), (46, 162), (48, 11), (49, 102), (50, 16), (50, 55), (50, 72), (50, 81), (50, 146), (51, 124), (52, 21), (54, 50), (57, 167), (59, 71), (61, 24), (62, 11), (63, 59), (64, 111), (65, 151), (67, 128), (70, 115), (72, 149), (74, 37), (75, 89), (76, 46), (77, 98), (78, 112), (80, 37), (83, 63), (85, 58), (87, 50), (88, 167), (89, 33), (90, 85), (91, 73), (93, 115), (96, 11), (98, 136), (100, 63), (101, 76), (102, 20), (103, 3), (103, 16), (103, 29), (103, 42), (103, 55), (103, 68), (103, 72), (103, 81), (103, 94), (103, 107), (103, 120), (103, 133), (103, 146), (103, 159), (104, 34), (109, 128), (111, 45), (113, 76), (114, 154), (115, 7), (116, 59), (117, 164), (119, 102), (122, 76), (124, 123), (126, 89), (127, 63), (128, 163), (129, 46), (130, 125), (132, 11), (135, 24), (137, 32), (140, 141), (141, 150), (142, 33), (143, 86), (145, 89), (148, 141), (150, 110), (153, 50), (154, 137), (155, 20), (156, 47), (158, 167), (161, 89), (163, 19), (165, 128), (166, 128), (167, 124), (168, 7)

p	k	B_p	classes (a, b)
13	3	3	(106, 869), (139, 2130), (152, 622), (275, 1883), (308, 102), (321, 791), (444, 700), (477, 271), (490, 960), (613, 1714), (646, 440), (659, 1129), (726, 42), (726, 211), (726, 380), (726, 549), (726, 718), (726, 887), (726, 1056), (726, 1225), (726, 1394), (726, 1563), (726, 1732), (726, 1901), (726, 2070), (782, 531), (815, 609), (828, 1298), (895, 120), (895, 289), (895, 458), (895, 627), (895, 796), (895, 965), (895, 1134), (895, 1303), (895, 1472), (895, 1641), (895, 1810), (895, 1979), (895, 2148), (951, 1545), (984, 778), (997, 1467), (1120, 362), (1153, 947), (1166, 1636), (1233, 3), (1233, 94), (1233, 172), (1233, 263), (1233, 341), (1233, 432), (1233, 510), (1233, 601), (1233, 679), (1233, 770), (1233, 848), (1233, 939), (1233, 1017), (1233, 1108), (1233, 1186), (1233, 1277), (1233, 1355), (1233, 1446), (1233, 1524), (1233, 1615), (1233, 1693), (1233, 1784), (1233, 1862), (1233, 1953), (1233, 2031), (1233, 2122), (1289, 1376), (1322, 1116), (1335, 1805), (1402, 133), (1402, 302), (1402, 471), (1402, 640), (1402, 809), (1402, 978), (1402, 1147), (1402, 1316), (1402, 1485), (1402, 1654), (1402, 1823), (1402, 1992), (1402, 2161), (1458, 193), (1491, 1285), (1504, 1974), (1571, 107), (1571, 159), (1571, 276), (1571, 328), (1571, 445), (1571, 497), (1571, 614), (1571, 666), (1571, 783), (1571, 835), (1571, 952), (1571, 1004), (1571, 1121), (1571, 1173), (1571, 1290), (1571, 1342), (1571, 1459), (1571, 1511), (1571, 1628), (1571, 1680), (1571, 1797), (1571, 1849), (1571, 1966), (1571, 2018), (1571, 2135), (1571, 2187), (1627, 1207), (1660, 1454), (1673, 2143), (1796, 24), (1829, 1623), (1842, 115), (1909, 29), (1909, 68), (1909, 198), (1909, 237), (1909, 367), (1909, 406), (1909, 536), (1909, 575), (1909, 705), (1909, 744), (1909, 874), (1909, 913), (1909, 1043), (1909, 1082), (1909, 1212), (1909, 1251), (1909, 1381), (1909, 1420), (1909, 1550), (1909, 1589), (1909, 1719), (1909, 1758), (1909, 1888), (1909, 1927), (1909, 2057), (1909, 2096), (1965, 1038), (1998, 1792), (2011, 284), (2134, 2052), (2167, 1961), (2180, 453)
17	1	0	(2, 3), (6, 14), (10, 14), (14, 10)
17	2	0	(1, 262), (7, 92), (8, 242), (9, 98), (12, 71), (15, 113), (15, 195), (15, 270), (18, 177), (24, 58), (25, 208), (26, 285), (29, 139), (32, 15), (32, 130), (32, 144), (35, 92), (41, 24), (42, 174), (43, 183), (46, 207), (49, 49), (49, 93), (49, 147), (52, 7), (58, 279), (59, 140), (60, 81), (63, 275), (66, 42), (66, 83), (66, 164), (69, 211), (75, 245), (76, 106), (77, 268), (80, 54), (83, 117), (83, 181), (83, 280), (86, 126), (92, 211), (93, 72), (94, 166), (97, 122), (100, 151), (100, 198), (100, 229), (103, 41), (109, 177), (110, 38), (111, 64), (114, 190), (117, 178), (117, 185), (117, 215), (120, 245), (126, 143), (127, 4), (128, 251), (131, 258), (134, 127), (134, 219), (134, 232), (137, 160), (143, 109), (144, 259), (145, 149), (148, 37), (151, 76), (151, 249), (151, 253), (154, 75), (160, 75), (161, 225), (162, 47), (165, 105), (168, 25), (168, 266), (168, 287), (171, 279), (177, 41), (178, 191), (179, 234), (182, 173), (185, 32), (185, 263), (185, 283), (188, 194), (194, 7), (195, 157), (196, 132), (199, 241), (202, 11), (202, 66), (202, 212), (205, 109), (211, 262), (212, 123), (213, 30), (216, 20), (219, 28), (219, 100), (219, 161), (222, 24), (228, 228), (229, 89), (230, 217), (233, 88), (236, 45), (236, 110), (236, 134), (239, 228), (245, 194), (246, 55), (247, 115), (250, 156), (253, 59), (253, 62), (253, 168), (256, 143), (262, 160), (263, 21), (264, 13), (267, 224), (270, 8), (270, 79), (270, 202), (273, 58), (279, 126), (280, 276), (281, 200), (284, 3), (287, 236), (287, 246)

p	k	B_p	classes (a, b)
17	3	0	(287, 3275), (576, 3564), (865, 3853), (1154, 4142), (1443, 4431), (1732, 4720), (2021, 96), (2310, 385), (2599, 674), (2888, 963), (3177, 1252), (3466, 1541), (3755, 1830), (4044, 2119), (4333, 2408), (4622, 2697), (4911, 2986)
29	1	0	(1, 15), (1, 19), (1, 24), (2, 6), (5, 22), (6, 10), (6, 14), (7, 23), (10, 25), (11, 2), (15, 13), (16, 8), (17, 8), (21, 2), (22, 6), (24, 23), (25, 13), (26, 18), (27, 8), (28, 25)
29	2	0	(6, 759), (13, 308), (19, 195), (35, 672), (42, 134), (48, 775), (64, 585), (71, 801), (77, 514), (93, 498), (100, 627), (106, 253), (122, 411), (129, 453), (135, 833), (151, 324), (158, 279), (164, 572), (180, 237), (187, 105), (193, 311), (209, 150), (216, 772), (222, 50), (238, 63), (245, 598), (251, 630), (267, 817), (274, 424), (280, 369), (296, 730), (303, 250), (309, 108), (325, 643), (332, 76), (338, 688), (354, 556), (361, 743), (367, 427), (383, 469), (390, 569), (396, 166), (412, 382), (419, 395), (425, 746), (441, 295), (448, 221), (454, 485), (470, 208), (477, 47), (483, 224), (499, 121), (506, 714), (512, 804), (528, 34), (535, 540), (541, 543), (557, 788), (564, 366), (570, 282), (586, 701), (593, 192), (599, 21), (615, 614), (622, 18), (628, 601), (644, 527), (651, 685), (657, 340), (673, 440), (680, 511), (686, 79), (702, 353), (709, 337), (715, 659), (731, 266), (738, 163), (744, 398), (760, 179), (767, 830), (773, 137), (789, 92), (796, 656), (802, 717), (818, 5), (825, 482), (831, 456)
31	1	0	(1, 5), (2, 9), (2, 25), (2, 28), (3, 26), (6, 12), (7, 18), (9, 19), (10, 4), (10, 10), (10, 17), (11, 4), (11, 10), (11, 17), (12, 3), (12, 6), (12, 22), (13, 12), (14, 12), (15, 9), (15, 25), (15, 28), (16, 2), (17, 8), (18, 2), (19, 18), (20, 19), (21, 11), (22, 9), (22, 25), (22, 28), (23, 9), (23, 25), (23, 28), (24, 0), (24, 1), (24, 30), (25, 23), (26, 20), (27, 3), (27, 6), (27, 22), (29, 12), (30, 4), (30, 10), (30, 17)
37	1	0	(0, 30), (2, 22), (3, 14), (5, 0), (5, 36), (6, 15), (8, 6), (8, 11), (8, 20), (9, 15), (10, 23), (12, 18), (15, 6), (15, 11), (15, 20), (17, 24), (18, 19), (19, 35), (21, 6), (21, 11), (21, 20), (24, 8), (24, 25), (25, 14), (26, 3), (27, 10), (27, 32), (30, 6), (30, 11), (30, 20), (31, 18), (32, 16)

p	k	B_p	classes (a, b)
37	2	0	(5, 297), (13, 696), (16, 52), (23, 423), (24, 670), (42, 445), (50, 844), (53, 1236), (60, 275), (61, 966), (79, 593), (87, 992), (90, 1051), (97, 127), (98, 1262), (116, 741), (124, 1140), (127, 866), (134, 1348), (135, 189), (153, 889), (161, 1288), (164, 681), (171, 1200), (172, 485), (190, 1037), (198, 67), (201, 496), (208, 1052), (209, 781), (227, 1185), (235, 215), (238, 311), (245, 904), (246, 1077), (264, 1333), (272, 363), (275, 126), (282, 756), (283, 4), (301, 112), (309, 511), (312, 1310), (319, 608), (320, 300), (338, 260), (346, 659), (349, 1125), (356, 460), (357, 596), (375, 408), (383, 807), (386, 940), (393, 312), (394, 892), (412, 556), (420, 955), (423, 755), (430, 164), (431, 1188), (449, 704), (457, 1103), (460, 570), (467, 16), (468, 115), (486, 852), (494, 1251), (497, 385), (504, 1237), (505, 411), (523, 1000), (531, 30), (534, 200), (541, 1089), (542, 707), (560, 1148), (568, 178), (571, 15), (578, 941), (579, 1003), (597, 1296), (605, 326), (608, 1199), (615, 793), (616, 1299), (634, 75), (642, 474), (645, 1014), (652, 645), (653, 226), (671, 223), (679, 622), (682, 829), (689, 497), (690, 522), (708, 371), (716, 770), (719, 644), (726, 349), (727, 818), (745, 519), (753, 918), (756, 459), (763, 201), (764, 1114), (782, 667), (790, 1066), (793, 274), (800, 53), (801, 41), (819, 815), (827, 1214), (830, 89), (837, 1274), (838, 337), (856, 963), (864, 1362), (867, 1273), (874, 1126), (875, 633), (893, 1111), (901, 141), (904, 1088), (911, 978), (912, 929), (930, 1259), (938, 289), (941, 903), (948, 830), (949, 1225), (967, 38), (975, 437), (978, 718), (985, 682), (986, 152), (1004, 186), (1012, 585), (1015, 533), (1022, 534), (1023, 448), (1041, 334), (1049, 733), (1052, 348), (1059, 386), (1060, 744), (1078, 482), (1086, 881), (1089, 163), (1096, 238), (1097, 1040), (1115, 630), (1123, 1029), (1126, 1347), (1133, 90), (1134, 1336), (1152, 778), (1160, 1177), (1163, 1162), (1170, 1311), (1171, 263), (1189, 926), (1197, 1325), (1200, 977), (1207, 1163), (1208, 559), (1226, 1074), (1234, 104), (1237, 792), (1244, 1015), (1245, 855), (1263, 1222), (1271, 252), (1274, 607), (1281, 867), (1282, 1151), (1300, 1), (1308, 400), (1311, 422), (1318, 719), (1319, 78), (1337, 149), (1345, 548), (1348, 237), (1355, 571), (1356, 374)
41	1	0	(0, 5), (2, 22), (6, 16), (7, 6), (8, 36), (10, 2), (10, 10), (10, 29), (11, 18), (11, 30), (11, 34), (12, 35), (15, 19), (16, 28), (17, 9), (20, 6), (22, 2), (23, 12), (23, 31), (23, 39), (25, 16), (26, 9), (27, 4), (28, 28), (29, 4), (31, 20), (32, 7), (32, 11), (32, 23), (33, 4), (34, 4), (35, 18), (35, 30), (35, 34), (37, 35), (39, 13), (40, 15)

p	k	B_p	classes (a, b)
41	2	0	(5, 1344), (22, 215), (22, 234), (46, 114), (63, 1013), (63, 1158), (87, 565), (104, 111), (104, 420), (128, 1016), (145, 890), (145, 1363), (169, 1467), (186, 625), (186, 1669), (210, 237), (227, 767), (227, 1568), (251, 688), (268, 830), (268, 1546), (292, 1139), (309, 92), (309, 644), (333, 1590), (350, 1035), (350, 1423), (374, 360), (391, 297), (391, 521), (415, 811), (432, 1240), (432, 1300), (456, 1262), (473, 398), (473, 502), (497, 32), (514, 1177), (514, 1445), (538, 483), (555, 275), (555, 707), (579, 934), (596, 1054), (596, 1650), (620, 1385), (637, 152), (637, 912), (661, 155), (678, 174), (678, 931), (702, 606), (719, 29), (719, 1117), (743, 1057), (760, 379), (760, 808), (784, 1508), (801, 1322), (801, 1587), (825, 278), (842, 584), (842, 685), (866, 729), (883, 1464), (883, 1527), (907, 1180), (924, 562), (924, 789), (948, 1631), (965, 51), (965, 1341), (989, 401), (1006, 439), (1006, 994), (1030, 852), (1047, 256), (1047, 1218), (1071, 1303), (1088, 316), (1088, 1199), (1112, 73), (1129, 461), (1129, 1095), (1153, 524), (1170, 193), (1170, 1404), (1194, 975), (1211, 666), (1211, 972), (1235, 1426), (1252, 70), (1252, 1609), (1276, 196), (1293, 849), (1293, 871), (1317, 647), (1334, 133), (1334, 1628), (1358, 1098), (1375, 726), (1375, 1076), (1399, 1549), (1416, 338), (1416, 1505), (1440, 319), (1457, 603), (1457, 1281), (1481, 770), (1498, 543), (1498, 1382), (1522, 1221), (1539, 480), (1539, 1486), (1563, 1672), (1580, 748), (1580, 1259), (1604, 442), (1621, 10), (1621, 357), (1645, 893), (1662, 953), (1662, 1136)
61	1	0	(2, 45), (3, 15), (3, 23), (4, 20), (5, 26), (6, 7), (7, 3), (8, 38), (8, 46), (9, 56), (10, 15), (10, 23), (11, 9), (11, 13), (11, 39), (13, 41), (17, 12), (17, 19), (17, 30), (18, 8), (18, 17), (18, 36), (19, 12), (19, 19), (19, 30), (20, 57), (21, 33), (22, 38), (22, 46), (25, 4), (27, 26), (28, 5), (30, 3), (32, 28), (33, 16), (34, 33), (35, 26), (36, 9), (36, 13), (36, 39), (39, 11), (42, 14), (44, 43), (47, 25), (47, 44), (47, 53), (48, 58), (51, 28), (53, 41), (55, 26), (56, 42), (56, 49), (58, 41), (59, 41), (60, 5)
61	2	0	(56, 2044), (117, 2776), (178, 3508), (239, 519), (300, 1251), (361, 1983), (422, 2715), (483, 3447), (544, 458), (605, 1190), (666, 1922), (727, 2654), (788, 3386), (849, 397), (910, 1129), (971, 1861), (1032, 2593), (1093, 3325), (1154, 336), (1215, 1068), (1276, 1800), (1337, 2532), (1398, 3264), (1459, 275), (1520, 1007), (1581, 1739), (1642, 2471), (1703, 3203), (1764, 214), (1825, 946), (1886, 1678), (1947, 2410), (2008, 3142), (2069, 153), (2130, 885), (2191, 1617), (2252, 2349), (2313, 3081), (2374, 92), (2435, 824), (2496, 1556), (2557, 2288), (2618, 3020), (2679, 31), (2740, 763), (2801, 1495), (2862, 2227), (2923, 2959), (2984, 3691), (3045, 702), (3106, 1434), (3167, 2166), (3228, 2898), (3289, 3630), (3350, 641), (3411, 1373), (3472, 2105), (3533, 2837), (3594, 3569), (3655, 580), (3716, 1312)